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# FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

CHE 104 — Past Questions & Materials

**by Renance**

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Course: CHE 104

Institution: Federal University of Technology, Akure

Sources in this pack: 10

Includes past questions and lecture notes

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## Paper 1 of 10 — 2nd SemesterCHE104Past QuestionsCHE 104 100 QUESTIONS BY PHOTOCHEM.pdf

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### QUESTIONS AND ANSWERS FOR CHE104

Based on the tutorial

#### Boiling Points & General Principles

1. Which of the following compounds has the highest boiling point?

- A. Ethyl chloride (35°C)
- B. Acetone (56°C)
- C. Heptane (98°C) (Correct)
- D. Hexane (69°C)

2. The boiling point of ethanol is approximately:

- A. 35°C
- B. 56°C
- C. 78°C (Correct)
- D. 98°C

3. Which compound boils at 80°C?

- A. Hexane
- B. Benzene (Correct)
- C. Ethanol
- D. Acetone

4. The boiling point of ethane is around:

- A. -89°C (Correct)
- B. 35°C
- C. 56°C
- D. 69°C

5. The importance of boiling points in organic chemistry includes:

- A. Identification of compounds (Correct)
- B. Changing melting points
- C. Determining solubility in water
- D. Predicting color of substances

6. No organic compound boils exactly at 100°C because:

- A. They decompose at 100°C
- B. Water alone boils at 100°C under normal conditions (Correct)

- C. Organic compounds are unstable
- D. All organic compounds are solids
7. Boiling points can differ by location due to changes in:
- A. Temperature
- B. Altitude and pressure (Correct)
- C. Density
- D. Purity
8. Something boils when:
- A. Vapour pressure equals atmospheric pressure (Correct)
- B. External pressure exceeds vapour pressure
- C. Density decreases
- D. Solubility increases
9. The calorimeter is used for:
- A. Measuring boiling points (Correct)
- B. Measuring density
- C. Filtering compounds
- D. Separating liquids
10. Which of the following is correct?
- A. All hydrocarbons are organic compounds, but not all organic compounds are hydrocarbons (Correct)
- B. All organic compounds are hydrocarbons
- C. Hydrocarbons contain oxygen
- D. Hydrocarbons are inorganic
11. Atmospheric pressure is directly proportional to:
- A. Density
- B. Boiling point (Correct)
- C. Solubility
- D. Polarity
12. Alcohols generally boil \_\_\_\_\_ hydrocarbons of similar carbon number.
- A. Slower than
- B. Faster than (Correct)
- C. Same as
- D. Less predictably than
13. Miscible liquids can be separated by:
- A. Steam distillation (Correct)

- B. Vacuum distillation
  - C. Filtration
  - D. Chromatography
14. Pressure-sensitive liquids are separated by:
- A. Gravity filtration
  - B. Pressure distillation (Correct)
  - C. Steam distillation
  - D. Sublimation
15. Two volatile liquids are best separated by:
- A. Vacuum distillation (Correct)
  - B. Chromatography
  - C. Filtration
  - D. Melting point test
16. A pycnometer is used to measure:
- A. Density of liquids (Correct)
  - B. Boiling point
  - C. Solubility
  - D. Temperatur
17. The most accurate test for impurities in a compound is:
- A. Boiling point test
  - B. Melting point test (Correct)
  - C. Distillation
  - D. Refractive index
18. The melting point of pure water is:
- A. 0°C (Correct)
  - B. 25°C
  - C. 32°C
  - D. 100°C
19. The strongest type of bond among the options is:
- A. Ionic bond (Correct)
  - B. Covalent bond
  - C. Hydrogen bond
  - D. Van der Waals
20. Which of the following is a covalent molecule?
- A. NaCl



B. H<sub>2</sub> (Correct)

C. CaCl<sub>2</sub>

D. KBr

Filtration, Sublimation & Drying Agents

21. Gravity filtration is most suitable for:

A. Cold solution

B. Warm/hot solution (Correct)

C. Volatile solution

D. Solid only

22. Vacuum filtration is suitable for:

A. Warm solution

B. Cold solution (Correct)

C. Volatile liquid

D. Miscible solution

23. Sublimation is the change of matter from:

A. Solid to liquid

B. Solid to gas (Correct)

C. Liquid to solid

D. Gas to liquid

24. The reverse process of sublimation is called:

A. Condensation

B. Deposition (Correct)

C. Evaporation

D. Melting

25. Which of the following is a common drying agent?

A. Sodium sulfate (Correct)

B. Sodium chloride

C. Potassium permanganate

D. Copper sulfate

26. Calcium chloride is used as a:

A. Oxidizing agent

B. Drying agent (Correct)

C. Reducing agent

D. Solvent

27. Silica gel is commonly used in packaging to:

- A. Absorb odors and moisture (Correct)
- B. Increase pressure
- C. Act as a preservative
- D. Generate heat

28. A desiccator is used to:

- A. Store moisture-sensitive compounds (Correct)
- B. Filter impurities
- C. Test melting points
- D. Measure density

29. Sodium sulfate as a drying agent works by:

- A. Absorbing water (Correct)
- B. Generating oxygen
- C. Neutralizing acids
- D. Reacting with carbon dioxide

30. Which drying agent prevents rusting in products?

- A. Silica gel (Correct)
- B. Sodium sulfate
- C. Calcium carbonate
- D. Zinc oxide

#### Chromatography

31. In thin layer chromatography, the retention factor ( $R_f$ ) is calculated as:

- A. Distance moved by solvent  $\div$  distance moved by compound
- B. Distance moved by compound  $\div$  distance moved by solvent (Correct)
- C. Solvent time  $\div$  distance moved
- D. Compound weight  $\div$  solvent volume

32. Examples of stationary phases in chromatography include:

- A. Sodium chloride and potassium nitrate
- B. Silica gel and alumina (Correct)
- C. Acetone and ethanol
- D. Hexane and benzene

33. Paper chromatography uses:

- A. Paper as stationary phase (Correct)
- B. Acetone as stationary phase
- C. Gas as stationary phase
- D. Glass beads as stationary phase

34. Activated charcoal is used to:
- A. Remove color from solutions (Correct)
  - B. Absorb water
  - C. Increase boiling point
  - D. Act as a reducing agent
35. Which chromatography technique uses silica plates?
- A. TLC (Thin Layer Chromatography) (Correct)
  - B. Gas chromatography
  - C. Paper chromatography
  - D. HPLC

#### Extraction & Separation

36. The instrument used for liquid-liquid extraction is:
- A. Separating funnel (Correct)
  - B. Desiccator
  - C. Pycnometer
  - D. Chromatograph
37. For funnel separation, organic compounds must be:
- A. Miscible with water
  - B. Immiscible with water (Correct)
  - C. Solid in form
  - D. Colorless
38. Miscible liquids can only be separated by:
- A. Distillation (Correct)
  - B. Filtration
  - C. Decantation
  - D. Crystallization
39. Steam distillation is used for:
- A. Heat-sensitive compounds (Correct)
  - B. Non-volatile solids
  - C. Metals
  - D. Miscible liquids
40. Vacuum distillation is used for:
- A. High-boiling liquids (Correct)
  - B. Soluble salts
  - C. Solids

D. Alcohols

#### Solubility & Bonding

41. Solubility follows the principle:

- A. Opposites dissolve opposites
- B. Like dissolves like (Correct)
- C. Solids dissolve in solids only
- D. Polar dissolves nonpolar

42. Hydrogen bonding is responsible for:

- A. Dissolving compounds in water (Correct)
- B. Lowering melting points
- C. Increasing molecular mass
- D. Changing color of compounds

43. Why don't hydrocarbons dissolve in water?

- A. They cannot form hydrogen bonds (Correct)
- B. They are too heavy
- C. They have strong ionic bonds
- D. They decompose

44. Ionic compounds dissolve well in:

- A. Nonpolar solvents
- B. Polar solvents like water (Correct)
- C. Hydrocarbons
- D. Alcohol-free solvents

45. Alcohols are soluble in water mainly due to:

- A. Ionic bonds
- B. Hydrogen bonding (Correct)
- C. Weak van der Waals forces
- D. Dipole-dipole repulsion

#### Instruments & Measurements

46. A refractometer is used to measure:

- A. Refractive index of compounds (Correct)
- B. Boiling points
- C. Melting points
- D. pH

47. The instrument that measures chirality of organic compounds is:

- A. Polarimeter (Correct)

B. Refractometer

C. Thermometer

D. Calorimeter

48. A thermometer is used to measure:

A. Temperature (Correct)

B. Boiling pressure

C. Mass

D. Density

49. Barometer measures:

A. Atmospheric pressure (Correct)

B. Boiling point

C. Mass of gases

D. Refractive index

50. Densities of liquids are measured using:

A. Pycnometer (Correct)

B. Calorimeter

C. Spectrophotometer

D. Thermometer

Pressure & Altitude

51. Pressure decreases with:

A. Higher altitude (Correct)

B. Lower altitude

C. Higher temperature

D. Increased volume

52. At high altitudes, boiling points of liquids are:

A. Higher

B. Lower (Correct)

C. Unchanged

D. Random

53. Atmospheric pressure is highest at:

A. Sea level (Correct)

B. High mountains

C. Space

D. Equator only

54. When pressure is reduced, boiling point:

- A. Increases
- B. Decreases (Correct)
- C. Stays constant
- D. Doubles

55. Vacuum distillation works by:

- A. Reducing pressure to lower boiling points (Correct)
- B. Increasing volume
- C. Adding solutes
- D. Cooling rapidly

Miscellaneous Concepts

56. The size of the sample does not affect:

- A. Melting point
- B. Boiling point (Correct)
- C. Density
- D. Refractive index

57. Impure substances generally have:

- A. Sharp melting points
- B. Lower and broader melting points (Correct)
- C. Higher boiling points always
- D. No effect on melting

58. "No B without BB" in lab slang means:

- A. No bread without Blue Band (margarine) (Correct)
- B. No benzene without boiling
- C. No base without buffer
- D. No bond without breaking

59. Hydrocarbons are made up of:

- A. Carbon and hydrogen only (Correct)
- B. Carbon and oxygen
- C. Hydrogen and nitrogen
- D. Oxygen and chlorine

60. Organic compounds may contain:

- A. Carbon only
- B. Carbon and hydrogen plus heteroatoms (Correct)
- C. Metals only
- D. Inorganic salts

Continuation (Q61 – Q100)

61. The boiling point of hexane is approximately:

- A. 69°C (Correct)
- B. 80°C
- C. 78°C
- D. 56°C

62. The boiling point of acetone is:

- A. 35°C
- B. 56°C (Correct)
- C. 69°C
- D. 80°C

63. The boiling point of ethyl chloride is:

- A. 25°C
- B. 35°C (Correct)
- C. 56°C
- D. 69°C

64. Among the following, which has the lowest boiling point?

- A. Benzene (80°C)
- B. Hexane (69°C)
- C. Acetone (56°C)
- D. Ethyl chloride (35°C) (Correct)

65. Which type of liquid is best separated by steam distillation?

- A. Heat-sensitive, immiscible liquids (Correct)
- B. High-boiling, miscible liquids
- C. Strong electrolytes
- D. Volatile metals

66. Boiling point is affected by:

- A. Intermolecular forces (Correct)
- B. Color of compound
- C. Shape of container
- D. Sample size

67. Hydrogen bonding is strongest in:

- A. Alcohols (Correct)
- B. Hydrocarbons
- C. Aldehydes
- D. Ethers

68. The bond formed by sharing electrons is:

- A. Covalent bond (Correct)
- B. Ionic bond
- C. Metallic bond
- D. Hydrogen bond

69. Ionic bonds are formed by:

- A. Transfer of electrons (Correct)
- B. Sharing of electrons
- C. Overlapping orbitals
- D. Hydrogen exchange

70. The change from liquid to gas is called:

- A. Condensation
- B. Evaporation/Boiling (Correct)
- C. Deposition
- D. Melting

#### □ Laboratory Techniques

71. The best test for purity of an organic solid is:

- A. Melting point (Correct)
- B. Boiling point
- C. Density
- D. Filtration

72. A sharp melting point indicates:

- A. High impurity
- B. High purity (Correct)
- C. Low density
- D. Low polarity

73. A broad melting range indicates:

- A. A pure compound
- B. An impure compound (Correct)
- C. A volatile liquid
- D. A hydrocarbon

74. In gravity filtration, which paper is commonly used?

- A. Whatman filter paper (Correct)
- B. Glossy paper
- C. Wax paper
- D. Tissue paper

75. In vacuum filtration, suction is provided by:

- A. Hand pump
- B. Water aspirator (Correct)
- C. Gas burner
- D. Desiccator

76. A Buchner funnel is used in:

- A. Vacuum filtration (Correct)
- B. Gravity filtration
- C. Chromatography
- D. Melting point determination



77. The solvent front in chromatography refers to:

- A. The furthest point reached by the solvent (Correct)
- B. The starting line
- C. The pigment line
- D. The  $R_f$  value

78.  $R_f$  values are always:

- A. Less than 1 (Correct)
- B. Greater than 1
- C. Equal to 1
- D. Negative

79. A desiccator usually contains:

- A. Drying agents (Correct)
- B. Boiling chips
- C. Solvents
- D. Lubricants

80. The separating funnel works on the principle of:

- A. Difference in solubility
- B. Difference in density of immiscible liquids (Correct)
- C. Difference in polarity
- D. Difference in color

#### □ Concepts in Physical Chemistry

81. The physical state directly changing from gas to solid is:

- A. Deposition (Correct)
- B. Sublimation
- C. Condensation
- D. Melting

82. Boiling point of a liquid is influenced by:

- A. Intermolecular forces and external pressure (Correct)
- B. Color of compound only
- C. Density only
- D. Melting point

83. The higher the molecular weight of hydrocarbons, the:

- A. Lower the boiling point
- B. Higher the boiling point (Correct)
- C. Constant boiling point
- D. Zero boiling point

84. Which compound among the following can form hydrogen bonds?

- A. Ethanol (Correct)
- B. Hexane
- C. Benzene
- D. Ethyl chloride

85. A nonpolar compound is best soluble in:

- A. Water
- B. Hexane (Correct)
- C. Ethanol
- D. Acetone

86. A polar compound is best soluble in:

- A. Benzene
- B. Water (Correct)
- C. Hexane
- D. Ether

87. When a liquid reaches its boiling point, its vapour pressure is:

- A. Equal to external pressure (Correct)
- B. Lower than external pressure
- C. Zero
- D. Higher than external pressure

88. The boiling point of a liquid at vacuum is:

- A. Lower than normal boiling point (Correct)
- B. Higher than normal boiling point
- C. Equal to 100°C
- D. Not measurable

89. The unit of refractive index is:

- A. No unit (dimensionless) (Correct)
- B. Joules
- C. Newtons
- D. Pascal

90. The melting point of ice is:

- A. 0°C (Correct)
- B. 25°C
- C. 100°C
- D. -10°C

□ Instruments & Miscellaneous

91. Which instrument measures optical activity?

- A. Polarimeter (Correct)
- B. Spectrometer
- C. Refractometer
- D. Thermometer

92. Which instrument is used in determining boiling point?

- A. Calorimeter (Correct)
- B. Polarimeter
- C. Pycnometer
- D. Chromatograph

93. Which of the following is NOT a drying agent?
- A. Silica gel
  - B. Calcium chloride
  - C. Sodium sulfate
  - D. Potassium permanganate (Correct)
94. Chromatography separates compounds based on:
- A. Polarity and solubility differences (Correct)
  - B. Color only
  - C. Size only
  - D. Mass only
95. A high  $R_f$  value means the compound is:
- A. More polar
  - B. Less polar (Correct)
  - C. Heavier
  - D. Impure
96. Which of the following hydrocarbons is an alkane?
- A. Hexane (Correct)
  - B. Benzene
  - C. Ethanol
  - D. Acetone
97. Which of the following is aromatic?
- A. Benzene (Correct)
  - B. Ethane
  - C. Ethanol
  - D. Hexane
98. Which functional group is present in ethanol?
- A. Hydroxyl group ( $-OH$ ) (Correct)
  - B. Carbonyl group ( $C=O$ )
  - C. Amino group ( $-NH_2$ )
  - D. Ether group ( $-O-$ )
99. Which functional group is present in acetone?
- A. Carbonyl group (ketone) (Correct)
  - B. Hydroxyl group
  - C. Carboxyl group
  - D. Amine group
100. Which of the following liquids is immiscible with water?
- A. Hexane (Correct)
  - B. Ethanol
  - C. Acetone
  - D. Ethyl alcohol

## Paper 2 of 10 — 2nd Semester CHE104 Past Questions CHE 104 POSSIBLE QUESTIONS AND ANSWERS COMPILED BY PHOTOC

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CHE 104 POSSIBLE QUESTIONS AND ANSWERS COMPILED BY

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CHE 104 QUESTIONS WITH SOLUTIONS

1. Which of the following compound is formed in the borax bead test? (a) Meta borate (b) Ortho borate (c ) Tetra borate (d) None of the above  
Answer: (a) Meta borate is formed in the borax bead test.

2. Which of the following metal does not give the borax bead test? (a) Chromium (b) Lead (c ) Silver (d) None of the above

Answer: Lead does not give the borax bead test.

3. Which of the following metal chloride imparts apple green colour to the Bunsen flame? (a) Sodium Chloride (b) Magnesium Chloride (c ) Calcium Chloride (d) None of the above

Answer: (b) Magnesium chloride imparts apple green colour to the bunsen flame.

4. Which of the following compound gets precipitated in detecting carbon and hydrogen? (a) Copper sulphate (b) Carbon dioxide (c) Calcium carbonate (d) None of the above

Answer: (c ) Calcium carbonate gets precipitated in the detection of carbon and hydrogen.

5. Which of the following element cannot be detected by Lassaigne's test? (a) Sulfur (b) Fluorine (c) Nitrogen (d) None of the above

Answer: (b) Fluorine cannot be detected by Lassaigne's test.

6. What is salt analysis?

Answer: Salt analysis, also known as qualitative inorganic or systematic qualitative analysis, is the qualitative and quantitative analysis of cations and anions present in an inorganic salt.

7. What is the preliminary test for chlorides?

Answer: Chlorides show the following characteristics 1. White colour 2. Intensification of pungent- smelling fumes when a glass rod dipped in ammonium hydroxide was brought near the mouth of the test tube

8. Which of the following elements can be identified by the Beilstein test? (a) Chlorine (b) Silver (c ) Nitrogen (d) None of the above

Answer: (a) Chlorine can be identified by using the Beilstein test.

9. The copper wire test for halogens is commonly known as (a) Lassaigne's Test (b) Duma's Test (c ) Beilstein's Test (d) Liebig's Test

Answer: (c ) The copper wire test for halogens is commonly known as Beilstein's Test.

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10. What is a preliminary test?

Answer: Preliminary test is performed first in the salt analysis. It is used for getting an indication of the radical.

11. Name the radicals that might be present in a white salt?

Answer: A white salt might contain a cuprous, ferrous, ferric, cobalt cation or a nickel cation in it.

12. What are acidic and basic radicals?

Answer: A radical containing a positive charge is known as a basic radical while a radical containing a negative charge is known as an acidic radical.

13. What is the colour of nickel salt?

Answer: Nickel salts are green or bluish-green in colour.

14. Name some salts that produce a crackling sound on heating.

Answer: Salts like lead nitrate, potassium bromide, barium nitrate, and sodium chloride produces a crackling sound on heating.

Practise Questions on Salt Analysis

1. Why is the need for salt analysis?

2. What is the primary principle of salt analysis?

3. What is the procedure for the chloride test?

4. What is the preliminary test of Group 3 anions?

5. What are the steps followed in the salt analysis?

A list of some common laboratory apparatus used in chemistry and their uses:

1. Beaker: used for holding and heating liquids.

2. Erlenmeyer flask: used for mixing, heating, and storing liquids. It has a narrow neck and a wider base to prevent spills.

3. Test tube: used for holding small amounts of liquids and observing chemical reactions.

4. Burette: used for precise measurement of liquids during titrations.

5. Pipette: used for precise measurement and transfer of small amounts of liquids.

6. Graduated cylinder: used for measuring the volume of liquids with high precision.

7. Funnel: used for transferring liquids from one container to another with minimal spillage.

8. Thermometer: used for measuring temperature.

9. Hot plate: used for heating substances.

10. Bunsen burner: used for heating, sterilizing, and burning substances.

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11. Separatory funnel: used for separating immiscible liquids or liquid-liquid extractions.

12. Distillation apparatus: used for separating the components of a mixture based on their boiling points.

13. Crucible: used for heating solids at high temperatures.

14. Mortar and pestle: used for grinding and mixing solids.

15. Spectrophotometer: used for measuring the absorption or transmission of light by a substance.

16. pH meter: used for measuring the acidity or alkalinity of a solution.

17. Conductivity meter: used for measuring the electrical conductivity of a solution.

18. Electrodes: used in electrochemical experiments to conduct electricity between the solution and the power source.

19. Gas syringe: used for measuring the volume of gases produced in a reaction.

20. Microscope: used for observing small particles or structures.

Read also on melting points of salts, types of salts, solubility of salts.

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1. What is a primary use of boiling point determination?

- a) Measuring density
- b) Identifying unknown compounds ☒
- c) Determining optical activity
- d) Calculating Rf values Answer

2. The presence of impurities in a liquid typically:

- a) Lowers the boiling point
- b) Raises the boiling point ☒
- c) Has no effect on the boiling point
- d) Creates an azeotrope

## ACADEMIC FOCUS

3. Why do alcohols have higher boiling points than hydrocarbons of similar molecular mass?

- a) Higher molecular weight
- b) Presence of hydrogen bonding ☒
- c) Increased polarity
- d) Lower vapor pressure

4. Which intermolecular force is associated with the lowest boiling points?

- a) Ionic bonding
- b) Hydrogen bonding
- c) Dipole-dipole forces
- d) Van der Waals forces ☒

5. Ether has a very low boiling point because it:

- a) Is highly polar
- b) Has a high molecular weight
- c) Lacks hydrogen bonding ☒
- d) Is soluble in water

6. The boiling point of a liquid is defined as the temperature where its:

- a) Density equals that of air
- b) Vapor pressure equals the external atmospheric pressure ☒
- c) Surface tension becomes zero
- d) Intermolecular forces are completely broken

7. At a high altitude, where atmospheric pressure is lower, the boiling point of water will:

- a) Increase
- b) Decrease ☒
- c) Remain the same
- d) Become unpredictable

8. A mixture of two immiscible liquids, like water and benzene, will boil:

- a) At the average of their individual boiling points
- b) At a temperature higher than either pure liquid
- c) At a temperature lower than either pure liquid ☒
- d) Only after one liquid has completely vaporized



9. Which technique exploits the principle in which two immiscible liquid boil at a lower temperature compared to their pure form.

- a) Vacuum distillation
- b) Simple distillation
- c) Steam distillation ☒
- d) Fractional distillation

10. Which compound would be expected to have the highest boiling point based on intermolecular forces? a)

- n-Hexane (MW ~86)
- b) Ethanol (MW ~46) ☒
- c) Chloroform (MW ~119)
- d) Diethyl ether (MW ~74)

11. An azeotrope is a mixture that:

- a) Boils over a wide temperature range
- b) Has a boiling point altered by impurities
- c) Boils at a constant temperature like a pure compound ☒
- d) Can only be separated by filtration

ACADEMIC FOCUS ; complied by Stephen (KAYJAY)  
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12. Which distillation method is most suitable for separating two liquids with boiling points of 78°C and

100°C?

- a) Steam distillation
- b) Vacuum distillation
- c) Simple distillation ☒
- d) Fractional distillation

13. Fractional distillation is preferred over simple distillation when the boiling points of the components are:

- a) Very different
- b) Identical
- c) Very close ☒
- d) Above 200°C

14. Vacuum distillation is used primarily for compounds that are:

- a) High-boiling or heat-sensitive ☒
- b) Highly volatile
- c) Immiscible with water
- d) Soluble in organic solvents

15. What is the purpose of a reflux condenser during heating?

- a) To increase the boiling point
- b) To prevent vapor loss ☒
- c) To remove colored impurities
- d) To measure the temperature accurately

16. Steam distillation is particularly useful for isolating:

- a) Inorganic salts
- b) High-boiling natural products like essential oils ☒
- c) Low-boiling petroleum ether
- d) Water-soluble sugars

17. The purity of an organic solid is best confirmed by its:

- a) Boiling point
- b) Density
- c) Melting point ☒
- d) Refractive index

ACADEMIC FOCUS

18. A pure solid is characterized by a melting point that is:

- a) Broad and low
- b) Sharp and fixed ☒
- c) Unmeasurable
- d) Higher than its boiling point

19. The presence of impurities in a solid sample causes the melting point to:

- a) Become sharp and higher
- b) Become broad and lower ☒
- c) Remain unchanged
- d) Become sharp and lower

20. What is the purpose of recrystallization?

- a) To separate liquid mixtures
- b) To purify solids ☒
- c) To determine boiling point
- d) To extract compounds from a mixture

21. An ideal recrystallization solvent should:

- a) Dissolve the compound when cold but not when hot
- b) Dissolve the compound when hot but not when cold ☒
- c) Not dissolve the compound at any temperature
- d) Be miscible with water

22. Activated charcoal is added during recrystallization to:

- a) Increase the boiling point
- b) Remove colored impurities ☒
- c) Act as a drying agent
- d) Speed up the crystallization

23. Gravity filtration is typically used for:

- a) Cold solutions to get large crystals
- b) Hot solutions to avoid premature crystallization ☒
- c) Separating immiscible liquids
- d) Drying crystals

24. Vacuum filtration is used for:

- a) Hot saturated solutions
- b) Rapid separation of crystals from a cold solution ☒
- c) Removing water from a wet organic solvent
- d) Measuring the volume of a liquid

### ACADEMIC FOCUS

25. Sublimation involves the direct transition from:

- a) Solid to liquid
- b) Liquid to gas
- c) Solid to vapor to solid ☒
- d) Gas to liquid

26. Which compound is best purified by sublimation? a)

- Sodium chloride
- b) Sucrose
- c) Naphthalene ☒
- d) Ethanol

27. Which is NOT a common drying agent?

- a) Anhydrous sodium sulfate
- b) Activated charcoal ☒
- c) Calcium chloride
- d) Silica gel

28. A desiccator is used to provide:

- a) A high-temperature environment
- b) A moisture-free environment ✓
- c) A vacuum for filtration
- d) A cold surface for condensation

29. How is the density of a liquid most accurately measured?

- a) Using a thermometer
- b) Using a pycnometer ✓
- c) Using a refractometer
- d) Using a separatory funnel

30. Specific gravity is defined as:

- a) Density of substance + density of water
- b) Density of substance - density of water
- c) Density of substance ÷ density of water ✓
- d) Density of water ÷ density of substance

31. Thin Layer Chromatography (TLC) is primarily used to:

- a) Determine boiling point
- b) Check the purity of a compound ✓
- c) Measure optical activity
- d) Dry organic solvents

32. In TLC, the stationary phase is typically:

- a) A solvent like hexane
- b) Silica gel or alumina ☒
- c) Water
- d) Iodine vapor

33. The  $R_f$  value is calculated as:

- a) Distance moved by solvent / distance moved by compound
- b) Distance moved by compound / distance moved by solvent ☒
- c) Distance moved by compound  $\times$  distance moved by solvent
- d) Distance moved by solvent - distance moved by compound

34. A high  $R_f$  value indicates the compound has:

- a) A high boiling point
- b) A strong interaction with the stationary phase
- c) A weak interaction with the stationary phase ☒
- d) A high molecular weight

35. Paper chromatography is effectively used to separate:

- a) Metal ions
- b) Pigments like those in spinach ☒
- c) Crude oil fractions
- d) Water from alcohol

## ACADEMIC FOCUS

36. A separatory funnel is used for:

- a) Distillation
- b) Recrystallization
- c) Extraction and separating immiscible liquids ☒
- d) Measuring melting points

37. For an extraction, the organic solvent used must be:

- a) Miscible with water
- b) Denser than water
- c) Immiscible with water ☒
- d) The same as the drying agent

38. Which solvent is immiscible with water and suitable for extraction?

- a) Acetone
- b) Ethanol
- c) Methanol
- d) Diethyl ether ☒

39. The principle "like dissolves like" explains that: a)

- Polar solutes dissolve in nonpolar solvents
- b) Nonpolar solutes dissolve in polar solvents
- c) Polar solutes dissolve in polar solvents ☒
- d) Solids always dissolve in liquids



40. Which class of compounds is soluble in water due to hydrogen bonding?

- a) hexane
- b) propanol ☒
- c) benzene
- d) chloroform

41. Why are carboxylic acids soluble in water?

- a) Due to their low molecular weight
- b) Due to hydrogen bonding ☒
- c) Because they are nonpolar
- d) Because they form salts instantly

### academic focus

42. As the carbon chain length increases in a homologous series, the solubility in water:

- a) Increases
- b) Decreases ☒
- c) Remains the same
- d) Becomes unpredictable

43. Which compound would be insoluble in water?

- a) Ethanol
- b) Acetic acid
- c) Sodium chloride
- d) n-Heptane ☒

44. Amines are soluble in water because they:

- a) Are highly nonpolar
- b) Can form hydrogen bonds ☒
- c) Have very high boiling points
- d) React with water to form acids

45. A solubility test can be useful for:

- a) Determining melting point
- b) Identifying functional groups ☒
- c) Measuring refractive index
- d) Performing vacuum distillation

### ACADEMIC FOCUS

46. Which instrument is used to measure optical activity?

- a) Refractometer
- b) Polarimeter ☒
- c) Pycnometer
- d) Ebulliometer

47. Which instrument is used to measure the refractive index of a liquid?

- a) Polarimeter
- b) Refractometer ☒
- c) Thermometer
- d) Hygrometer

48. For an accurate boiling point measurement, the thermometer bulb should be placed:

- a) In the liquid itself
- b) In the vapor just above the liquid ☒
- c) On the outside of the apparatus
- d) In a cold water bath

49. Which method is NOT used for boiling point determination?

- a) Distillation setup
- b) Thiele tube method
- c) Vacuum filtration ☒
- d) Capillary tube method

50. A drying oven is commonly used to:

- a) Heat reactions under reflux
- b) Dry glassware and solid samples ☒
- c) Perform distillations
- d) Measure the density of liquids

51. which compound has the lowest boiling point?

- a) Acetone ☒
- b) Chloroform
- c) n-Hexane
- d) Ethanol

52. Which two compounds have very similar boiling points?

- a) Ethanol and Benzene ☒
- b) Chloroform and Acetone
- c) n-Hexane and n-Heptane
- d) Water and Ethanol

53. Organic solvents generally have lower boiling points than water because they:

- a) Have higher molecular weights
- b) Lack strong hydrogen bonding networks ☒
- c) Are more polar
- d) Are always impure

## academic focus

54. Which technique separates liquids based on differences in their boiling points?

- a) Recrystallization
- b) Extraction
- c) Distillation ☒
- d) Chromatography

55. Which pair of techniques is used specifically for purifying solids?

- a) Distillation and Extraction
- b) Recrystallization and Sublimation ☒
- c) Chromatography and Refraction
- d) Filtration and Reflux

56. The principle of partition between a stationary and a mobile phase is key to:

- a) Distillation
- b) Filtration
- c) Chromatography ☒
- d) Solubility testing

57. After an extraction with a wet organic solvent, the next step is to add:

- a) More water
- b) A drying agent ☒
- c) Activated charcoal
- d) A second immiscible solvent

58. Which factor does NOT affect the boiling point of a liquid?

- a) Atmospheric pressure
- b) Intermolecular forces
- c) The color of the liquid ☒
- d) Purity

59. The step-by process of dissolving a solid in hot solvent, filtering, and cooling to form pure crystals is called:

- a) Sublimation
- b) Distillation

- c) Extraction
- d) Recrystallization ☒

60. The overall goal of most practical techniques like recrystallization, distillation, is:

- a) To increase the yield of a reaction
- b) To identify and purify substances ☒
- c) To measure physical constants accurately
- d) To synthesize new compounds

61. Which is a better solvent for crystallization ;

- A) Benzene soluble when hot , insoluble when cold ☒
- B) Ethanol soluble when hot , soluble when cold
- C) Toluene insoluble when hot, insoluble when cold
- D) Fluorine insoluble when hot, soluble when cold.

62. Organic solvent generally have boiling point lesser than water. True or False

True ☒

63. The boiling point of chloroform is

- a) 78°
- b) 48°
- c) 61° ☒
- d) 105°

63. A liquid boils

- a) when atmospheric pressure is twice the vapor

Academic Focus

pressure

- b) when the atmospheric pressure is at STP
- c) when vapor pressure is higher than atmospheric pressure
- d) when vapor pressure equals atmospheric pressure ☒

64. What can be used in place of a hydrometer

- a) hygrometer
- b) barometer
- c) thermometer
- d) pycnometer ☒

65. When a gas converts directly to solid it is referred to as

- a) decomposition
- b) condensation
- c) deposition ☒
- d) compilation

66. What is the accurate method of determining boiling point ?

- a) capillarity method
- b) distillation method
- c) ebuillimeter
- d) all of the above ☒

67. When water is used to break a bond we refer to it as

- a) hydration
- b) dehydration
- c) hydrogenation
- d) hydrolysis ☒

68. Solubility test can distinguish between \_\_\_\_

- a) solids and liquids
- b) acid and base
- c) polar and non-polar ☒
- d) alkanols and esters

69. Which substance is least soluble in water

- a) propanol
- b) ethanol
- c) heptanol ☒
- d) methanol

## ACADEMIC FOCUS

compiled by  
Stephen (KAYJAY)  
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## Paper 4 of 10 — BAMBOY 100L 2nd semCHE 104 (BAMBOY)CHE 104 POSSIBLE QUESTIONS AND ANSWERS by MCCOY(TGM).pd

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### CHE 104 POSSIBLE QUESTIONS AND ANSWERS COMPILED BY PSALMMY MOVEMENT GROUP.

- CHE104 QUESTIONS WITH SOLUTIONS
1. Which of the following compound is formed in the borax bead test? (a) Meta borate (b) Ortho borate (c ) Tetra borate (d) None of the above  
Answer: (a) Meta borate is formed in the borax bead test.
2. Which of the following metal does not give the borax bead test? (a) Chromium (b) Lead (c ) Silver (d) None of the above  
Answer: Lead does not give the borax bead test.
3. Which of the following metal chloride imparts apple green colour to the Bunsen flame? (a) Sodium Chloride (b) Magnesium Chloride (c ) Calcium Chloride (d) None of the above  
Answer: (b) Magnesium chloride imparts apple green colour to the bunsen flame.
4. Which of the following compound gets precipitated in detecting carbon and hydrogen? (a) Copper sulphate (b) Carbon dioxide (c ) Calcium carbonate (d) None of the above  
Answer: (c ) Calcium carbonate gets precipitated in the detection of carbon and hydrogen.
5. Which of the following element cannot be detected by Lassaigne's test? (a) Sulfur (b) Fluorine (c) Nitrogen (d) None of the above  
Answer: (b) Fluorine cannot be detected by Lassaigne's test.
6. What is salt analysis?  
Answer: Salt analysis, also known as qualitative inorganic or systematic qualitative analysis, is the qualitative and quantitative analysis of cations and anions present in an inorganic salt.
7. What is the preliminary test for chlorides?  
Answer: Chlorides show the following characteristics 1. White colour 2. Intensification of pungent- smelling fumes when a glass rod dipped in ammonium hydroxide was brought near the mouth of the test tube
8. Which of the following elements can be identified by the Beilstein test? (a) Chlorine (b) Silver (c ) Nitrogen (d) None of the above  
Answer: (a) Chlorine can be identified by using the Beilstein test.
9. The copper wire test for halogens is commonly known as (a) Lassaigne's Test (b) Duma's Test (c ) Beilstein's Test (d) Liebig's Test  
Answer: (c ) The copper wire test for halogens is commonly known as Beilstein's Test.

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10. What is a preliminary test?

Answer: Preliminary test is performed first in the salt analysis. It is used for getting an indication of the radical.

11. Name the radicals that might be present in a white salt?

Answer: A white salt might contain a cuprous, ferrous, ferric, cobalt cation or a nickel cation in it.

12. What are acidic and basic radicals?

Answer: A radical containing a positive charge is known as a basic radical while a radical containing a negative charge is known as an acidic radical.

13. What is the colour of nickel salt?

Answer: Nickel salts are green or bluish-green in colour.

14. Name some salts that produce a crackling sound on heating.

Answer: Salts like lead nitrate, potassium bromide, barium nitrate, and sodium chloride produces a crackling sound on heating.

Practise Questions on Salt Analysis

1. Why is the need for salt analysis?

2. What is the primary principle of salt analysis?

3. What is the procedure for the chloride test?

4. What is the preliminary test of Group 3 anions?

5. What are the steps followed in the salt analysis?

A list of some common laboratory apparatus used in chemistry and their uses:

1. Beaker: used for holding and heating liquids.

2. Erlenmeyer flask: used for mixing, heating, and storing liquids. It has a narrow neck and a wider base to prevent spills.

3. Test tube: used for holding small amounts of liquids and observing chemical reactions.

4. Burette: used for precise measurement of liquids during titrations.

5. Pipette: used for precise measurement and transfer of small amounts of liquids.

6. Graduated cylinder: used for measuring the volume of liquids with high precision.

7. Funnel: used for transferring liquids from one container to another with minimal spillage.

8. Thermometer: used for measuring temperature.

9. Hot plate: used for heating substances.

10. Bunsen burner: used for heating, sterilizing, and burning substances.

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11. Separatory funnel: used for separating immiscible liquids or liquid-liquid extractions.
12. Distillation apparatus: used for separating the components of a mixture based on their boiling points.
13. Crucible: used for heating solids at high temperatures.
14. Mortar and pestle: used for grinding and mixing solids.
15. Spectrophotometer: used for measuring the absorption or transmission of light by a substance.
16. pH meter: used for measuring the acidity or alkalinity of a solution.
17. Conductivity meter: used for measuring the electrical conductivity of a solution.
18. Electrodes: used in electrochemical experiments to conduct electricity between the solution and the power source.
19. Gas syringe: used for measuring the volume of gases produced in a reaction.
20. Microscope: used for observing small particles or structures.

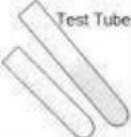
Read also on melting points of salts, types of salts, solubility of salts.

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## Lab Apparatus List for Student Use

| Description                                                                | Apparatus                                                                                            | Use                                                                                   | Description                                                                                    | Apparatus                                                                                             | Use                                                                                      |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| glass<br>100 mL<br>250 mL<br>400 mL<br>marked on the beaker                |  Beaker             | used as a container, like a cup, may be heated                                        | small porcelain dish with cover                                                                |  Crucible and Cover | used to heat small amount of solid material at high temperature                          |
| glass<br>several sizes                                                     |  Test Tube          | many uses<br>can be heated                                                            | alcohol as the fuel                                                                            |  Alcohol Burner     | Used to heat chemicals in beaker or test tube                                            |
| glass<br>marked with a milliliter (mL) scale                               |  Graduated Cylinder | used to measure volume                                                                | glass<br>marked with a milliliter scale and fitted with a stopcock, pinch clamp, or glass bead |  Buret             | used in the preparation of solutions                                                     |
| glass<br>common sizes<br>125 mL<br>250 mL<br>500 mL<br>marked on the flask |  Flask             | may be heated                                                                         | metal                                                                                          |  Tweezer          | used to pick up small objects or pulling out hairs or splinters                          |
| metal                                                                      |  Tongs            | used to pick up or hold apparatus                                                     | glass, rubber                                                                                  |  Dropper          | used to transfer small amounts of liquid                                                 |
| porcelain                                                                  |  Evaporation Dish | used as a container for small amount of liquid being evaporated                       | curved glass                                                                                   |  Watch Glass      | may be used as a beaker cover<br>may be used in evaporating very small amounts of liquid |
| glass or plastic                                                           |  Funnel           | used to pour liquids into containers which have a small opening, for example, bottles |                                                                                                |                                                                                                       |                                                                                          |

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# Paper 5 of 10 — BAMBOY 100L 2nd sem CHE 104 (BAMBOY) CHE 104 Tutorial .pdf

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|                                                                                                                                                                                                                                                  |                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| CHE 104 TUTORIAL                                                                                                                                                                                                                                 | § Separation of mixtures.                                                        |
| Acid dissolves in base                                                                                                                                                                                                                           | When impurities are present in samples it increases the boiling point.           |
| Diprotic dissolves in both Acid and base.                                                                                                                                                                                                        | A substance boils when the internal pressure equals the external pressure.       |
| Diagram of Laboratory Apparatus                                                                                                                                                                                                                  | Organic compound                                                                 |
| Observation, test and inferences                                                                                                                                                                                                                 | Boiling point                                                                    |
| Touch chem 103 practical                                                                                                                                                                                                                         | n-heptane 98°C                                                                   |
| Organic solvents have boiling point lower than that of water because the inter-molecular forces binding them together are weak. The longer the chain the higher their boiling point. Water has a strong force [Hydrogen bonding, ionic bonding]. | Ether 35°C                                                                       |
| ↓<br>Oxygen of a particular molecule                                                                                                                                                                                                             | Benzene 80°C                                                                     |
| Water bonds with hydrogen of another molecule of water.                                                                                                                                                                                          | 1                                                                                |
| INSTRUMENT USED TO MEASURE BOILING POINT OF ORGANIC SOLVENT is thermometer.                                                                                                                                                                      | Ethanol 78°C                                                                     |
| The bob of the thermometer is deeped inside the organic solvent without touching the walls of the container.                                                                                                                                     | Acetone 56°C                                                                     |
| USE OF BOILING POINT                                                                                                                                                                                                                             | n-hexane 69°C                                                                    |
| 1) To identify organic compounds (pure compounds have low boiling point)                                                                                                                                                                         | Chloroform 61°C                                                                  |
| 2) for purity check.                                                                                                                                                                                                                             | Organic                                                                          |
|                                                                                                                                                                                                                                                  | Ether has a lower boiling point lower compared to water.                         |
|                                                                                                                                                                                                                                                  | Mixture of two immiscible liquid usually boils at different temperature.         |
|                                                                                                                                                                                                                                                  | Impurities increases the boiling point.                                          |
|                                                                                                                                                                                                                                                  | Which organic compound boils at 100°C as water?                                  |
|                                                                                                                                                                                                                                                  | Ans: None of the above.                                                          |
|                                                                                                                                                                                                                                                  | Boiling point may differ per location because of different atmospheric pressure. |
|                                                                                                                                                                                                                                                  | Hydrocarbon - Hydrogen and carbon only.                                          |

|                                                                                                                                |                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| When atmospheric pressure decreases the boiling point decreases.                                                               | Crystallization: To purify solids.                                                                                           |
| In the family of organic compounds Alcohols boil higher than hydrocarbons of similar molecular weight.                         | Impurities should be insoluble and filtered out. Water, ethanol and acetone are for recovering solvents from common solvent. |
| To separate liquids with similar boiling point we use fractional distillation.                                                 | Benzene for recrystallization.                                                                                               |
| Steam distillation are used for immiscible liquid.                                                                             | Filtration                                                                                                                   |
| Vacuum distillation (lower pressure condition) is used to separate mixtures that are heat sensitive.                           | → Gravity and vacuum distillation.                                                                                           |
| DENSITY AND PURITY TEST                                                                                                        | ↓<br>To filter hot solution      To filter cold solution                                                                     |
| Instrument used to measure density is hydrometer. Pyrometer                                                                    | The opposite of sublimation is deposition.                                                                                   |
| Specific Gravity = $\frac{\text{Density of Substance}}{\text{Density of water}}$                                               | Sublimation is used to purify volatile solids like Iodine and Amine.                                                         |
| The most accurate way of checking purity of a substance is the melting point because most times they have sharp melting point. | Dry agent or desiccator.                                                                                                     |
| Pure solids have sharp melting point.                                                                                          | → Silica gel                                                                                                                 |
| Impure solids melting point is over a range.                                                                                   | → Calcium chloride (liquid to liquid absorbent).                                                                             |
| Does water melt                                                                                                                | → Anhydrous sodium sulphate                                                                                                  |
|                                                                                                                                | Chromatography                                                                                                               |
|                                                                                                                                | Thin layer chromatography                                                                                                    |
|                                                                                                                                | $R_F = \frac{\text{Distance moved by compound}}{\text{Distance moved by solvent}}$                                           |
|                                                                                                                                | High $R_F$ value = weak interaction with stationary phase                                                                    |
|                                                                                                                                | Paper chromatography separates pigments                                                                                      |
|                                                                                                                                | liquid solvent      stationary phase (silica gel / alumina)                                                                  |
|                                                                                                                                | The principle behind TLC is partition between mobile phase and stationary phase                                              |
|                                                                                                                                | <del>stationary phase and solvent</del> <del>solvent</del>                                                                   |

To remove coloured impurities during crystallization, you use activated charcoal.

### EXTRACTION [Separating funnel method]

used to separate immiscible liquids.

Solvent must be immiscible with water.

Solvent include ether

Polarimeter  $\rightarrow$  measure refractive index

Polymerimeter is used to measure optical

activity of organic compounds.

Hexane is insoluble in water because

there is no  $-OH$  (hydrogen bond).

All hydrocarbons cannot form hydrogen bonds.

Electron pull will go towards the more electronegative

side.

Polar solute dissolves in a polar solvent

non-polar solute will dissolve in a non-polar

solvent.

Chromatography can be used to check

purity.

~~Pressure~~ Pressure above sea level: 760 mmHg

$\therefore$  Water boils at pressure above sea level.

Pressure is lower

At higher altitude, boiling point decreases

An abnormal low boiling point indicates weak intermolecular forces

There is only one factor that does not affect boiling point and that is size of sample

## Paper 6 of 10 — BAMBOY 100L 2nd semCHE 104 (BAMBOY)CHE104 PRACTICE QUESTION AND ANSWER - BIG BOSS.pdf

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### CHE 104 POSSIBLE QUESTIONS AND ANSWERS - BIG BOSS

1. Which of the following compounds has the highest boiling point?

- A. Ethyl chloride (35°C)
- B. Acetone (56°C)
- C. Heptane (98°C) +++
- D. Hexane (69°C)

Explanation: Longer carbon chain increases van der Waals forces, giving heptane the highest boiling points

2. The boiling point of ethanol is approximately:

- A. 35°C
- B. 56°C
- C. 78°C +++
- D. 98°C

Explanation: Ethanol boils at about 78°C at standard pressure.

3. Which compound boils at 80°C?

- A. Hexane
- B. Benzene +++
- C. Ethanol
- D. Acetone

Explanation: Benzene has a boiling point close to 80°C

4. The boiling point of ethane is around:

- A. -89°C +++
- B. 35°C
- C. 56°C
- D. 69°C

Explanation: Ethane is a simple alkane with very low boiling point.



5. The importance of boiling points in organic chemistry includes:

- A. Identification of compounds +++
- B. Changing melting points
- C. Determining solubility in water
- D. Predicting color of substances

Explanation: Boiling points are physical properties used to identify and check purity.

6. No organic compound boils exactly at 100°C because:

- A. They decompose at 100°C
- B. Water alone boils at 100°C under normal conditions +++
- C. Organic compounds are unstable
- D. All organic compounds are solids

Explanation: 100°C is the characteristic boiling point of water at 1 atm.

7. Boiling points can differ by location due to changes in:

- A. Temperature
- B. Altitude and pressure +++
- C. Density
- D. Purity

Explanation: Atmospheric pressure varies with altitude, affecting boiling points.

8. Something boils when:

- A. Vapour pressure equals atmospheric pressure +++
- B. External pressure exceeds vapour pressure
- C. Density decreases
- D. Solubility increases

Explanation: Boiling starts when vapour pressure = external pressure.

9. The calorimeter is used for:

- A. Measuring boiling points +++
- B. Measuring density

Explanation: Calorimeters measure heat-related properties, including boiling points.

10. Which of the following is correct?

- A. All hydrocarbons are organic compounds, but not all organic compounds are hydrocarbons +++
- B. All organic compounds are hydrocarbons
- C. Hydrocarbons contain oxygen
- D. Hydrocarbons are inorganic

Explanation: Hydrocarbons contain only C and H; organic compounds may contain other atoms.

11. Atmospheric pressure is directly proportional to:

- A. Density
- B. Boiling point +++
- C. Solubility
- D. Polarity

Explanation: Higher atmospheric pressure raises boiling point.

12. Alcohols generally boil \_\_\_\_ hydrocarbons of similar carbon number.

- A. Slower than
- B. Faster than +++
- C. Same as
- D. Less predictably than

Explanation: Hydrogen bonding makes alcohols have higher boiling points.

13. Miscible liquids can be separated by:

- A. Steam distillation +++

B. Vacuum distillation

C. Filtration

D. Chromatography

Explanation: Steam distillation separates miscible liquids based on boiling points.

14. Pressure-sensitive liquids are separated by:

A. Gravity filtration

B. Pressure distillation +++

C. Steam distillation

D. Sublimation

Explanation: Pressure distillation prevents decomposition by lowering boiling point.

15. Two volatile liquids are best separated by:

A. Vacuum distillation +++

B. Chromatography

C. Filtration

D. Melting point test

Explanation: Vacuum distillation reduces boiling points for separation.

16. A pycnometer is used to measure:

A. Density of liquids +++

B. Boiling point

C. Solubility

D. Temperature

Explanation: Pycnometers give precise liquid density.

17. The most accurate test for impurities in a compound is:

A. Boiling point test

B. Melting point test +++

C. Distillation

D. Refractive index

Explanation: Impurities lower and broaden melting point, making it a good purity test.

18. The melting point of pure water is:

A. 0°C +++

B. 25°C

C. 32°C

D. 100°C

Explanation: Water freezes and melts at 0°C at 1 atm.

19. The strongest type of bond among the options is:

A. Ionic bond +++

B. Covalent bond

C. Hydrogen bond

D. Van der Waals

Explanation: Ionic bonds are stronger than covalent and intermolecular forces.

20. Which of the following is a covalent molecule?

A. NaCl

B. H<sub>2</sub> +++

C. CaCl<sub>2</sub>

D. KBr

Explanation: H<sub>2</sub> is held by covalent bonds between two hydrogen atoms.

21. Gravity filtration is most suitable for:

A. Cold solution

B. Warm/hot solution +++

C. Volatile solution

D. Solid only

Explanation: Hot solutions prevent crystallization during filtration.

22. Vacuum filtration is suitable for:

A. Warm solution

B. Cold solution +++

C. Volatile liquid

D. Miscible solution

Explanation: Cold solutions are filtered quickly by vacuum.

23. Sublimation is the change of matter from:

A. Solid to liquid

B. Solid to gas +++

C. Liquid to solid

D. Gas to liquid

Explanation: Sublimation skips the liquid phase.

24. The reverse process of sublimation is called:

A. Condensation

B. Deposition +++

C. Evaporation

D. Melting

Explanation: Deposition is gas → solid.

25. Which of the following is a common drying agent?

A. Sodium sulfate +++

B. Sodium chloride

C. Potassium permanganate

D. Copper sulfate

Explanation: Sodium sulfate absorbs water effectively.

26. Calcium chloride is used as a:

- A. Oxidizing agent
- B. Drying agent +++
- C. Reducing agent
- D. Solvent

Explanation: Calcium chloride is commonly used as a desiccant.

27. Silica gel is commonly used in packaging to:

- A. Absorb odors and moisture +++
- B. Increase pressure
- C. Act as a preservative
- D. Generate heat

Explanation: Silica gel prevents moisture damage.

28. A desiccator is used to:

- A. Store moisture-sensitive compounds +++
- B. Filter impurities
- C. Test melting points
- D. Measure density

Explanation: Desiccators maintain dryness of compounds.

29. Sodium sulfate as a drying agent works by:

- A. Absorbing water +++
- B. Generating oxygen
- C. Neutralizing acids
- D. Reacting with carbon dioxide

Explanation: Sodium sulfate binds water molecules.

30. Which drying agent prevents rusting in products?

- A. Silica gel +++
- B. Sodium sulfate
- C. Calcium carbonate
- D. Zinc oxide

Explanation: Silica gel reduces rust by removing moisture.

31. In thin layer chromatography, the retention factor ( $R_f$ ) is calculated as:

- A. Distance moved by solvent  $\div$  distance moved by compound
- B. Distance moved by compound  $\div$  distance moved by solvent +++
- C. Solvent time  $\div$  distance moved
- D. Compound weight  $\div$  solvent volume

Explanation:  $R_f = \text{distance compound moved} \div \text{distance solvent moved}$ .

32. Examples of stationary phases in chromatography include:

- A. Sodium chloride and potassium nitrate
- B. Silica gel and alumina +++
- C. Acetone and ethanol
- D. Hexane and benzene

Explanation: Silica gel and alumina are common adsorbents.

33. Paper chromatography uses:

- A. Paper as stationary phase +++
- B. Acetone as stationary phase
- C. Gas as stationary phase
- D. Glass beads as stationary phase

Explanation: Paper holds water molecules, acting as stationary phase.

34. Activated charcoal is used to:

- A. Remove color from solutions +++
- B. Absorb water
- C. Increase boiling point
- D. Act as a reducing agent

Explanation: Charcoal adsorbs colored impurities.

35. Which chromatography technique uses silica plates?

- A. TLC (Thin Layer Chromatography) +++
- B. Gas chromatography
- C. Paper chromatography
- D. HPLC

Explanation: TLC plates are coated with silica gel.

36. The instrument used for liquid-liquid extraction is:

- A. Separating funnel +++
- B. Desiccator
- C. Pycnometer

Explanation: A separating funnel separates immiscible liquids.

37. For funnel separation, organic compounds must be:

- A. Miscible with water
- B. Immiscible with water +++
- C. Solid in form
- D. Colorless

Explanation: Immiscibility ensures phase separation.



38. Miscible liquids can only be separated by:

- A. Distillation +++
- B. Filtration
- C. Decantation
- D. Crystallization

Explanation: Distillation exploits boiling point differences.

39. Steam distillation is used for:

- A. Heat-sensitive compounds +++
- B. Non-volatile solids
- C. Metals
- D. Miscible liquids

Explanation: Steam allows distillation at lower temperatures.

40. Vacuum distillation is used for:

- A. High-boiling liquids +++
- B. Soluble salts
- C. Solids
- D. Alcohols

Explanation: Reduces boiling point of high-boiling compounds.

41. Solubility follows the principle:

- A. Opposites dissolve opposites
- B. Like dissolves like +++
- C. Solids dissolve in solids only
- D. Polar dissolves nonpolar

Explanation: Polar dissolves polar; nonpolar dissolves nonpolar.

42. Hydrogen bonding is responsible for:

- A. Dissolving compounds in water +++
- B. Lowering melting points
- C. Increasing molecular mass
- D. Changing color of compounds

Explanation: Hydrogen bonds allow solubility in polar solvents.

43. Why don't hydrocarbons dissolve in water?

- A. They cannot form hydrogen bonds +++
- B. They are too heavy
- C. They have strong ionic bonds
- D. They decompose

Explanation: Hydrocarbons are nonpolar, lacking H-bonding ability.

44. Ionic compounds dissolve well in:

- A. Nonpolar solvents
- B. Polar solvents like water +++
- C. Hydrocarbons
- D. Alcohol-free solvents

Explanation: Strong ionic interactions are stabilized by polar solvents.

45. Alcohols are soluble in water mainly due to:

- A. Ionic bonds
- B. Hydrogen bonding +++
- C. Weak van der Waals forces

Explanation: OH group forms hydrogen bonds with water.

46. A refractometer is used to measure:

A. Refractive index of compounds +++

B. Boiling points

C. Melting points

D. pH

Explanation: Refractometers determine refractive index.

47. The instrument that measures chirality of organic compounds is:

A. Polarimeter +++

B. Refractometer

C. Thermometer

D. Calorimeter

Explanation: Polarimeter measures optical activity of chiral molecules.

48. A thermometer is used to measure:

A. Temperature +++

B. Boiling pressure

C. Mass

D. Density

Explanation: Thermometers directly read temperature.

49. Barometer measures:

A. Atmospheric pressure +++

B. Boiling point

C. Mass of gases

D. Refractive index

Explanation: Barometer measures air pressure.

50. Densities of liquids are measured using:

A. Pycnometer +++

- B. Calorimeter
- C. Spectrophotometer
- D. Thermometer

Explanation: Pycnometer gives precise density values.

51. Pressure decreases with:

- A. Higher altitude +++
- B. Lower altitude
- C. Higher temperature
- D. Increased volume

Explanation: Air pressure drops with altitude.

52. At high altitudes, boiling points of liquids are:

- A. Higher
- B. Lower +++
- C. Unchanged
- D. Random

Explanation: Low pressure lowers boiling points.

53. Atmospheric pressure is highest at:

- A. Sea level +++
- B. High mountains
- C. Space
- D. Equator only

Explanation: Maximum pressure occurs at sea level.

54. When pressure is reduced, boiling point:

- A. Increases
- B. Decreases +++

C. Stays constant

D. Doubles

Explanation: Lower external pressure lowers boiling point.

55. Vacuum distillation works by:

A. Reducing pressure to lower boiling points +++

B. Increasing volume

C. Adding solutes

D. Cooling rapidly

Explanation: Reduced pressure allows high-boiling compounds to distill.

56. The size of the sample does not affect:

A. Melting point

B. Boiling point +++

C. Density

D. Refractive index

Explanation: Boiling point is an intrinsic property.

57. Impure substances generally have:

A. Sharp melting points

B. Lower and broader melting points +++

C. Higher boiling points always

D. No effect on melting

Explanation: Impurities disrupt crystal lattice, broadening melting range.

58. "No B without BB" in lab slang means:

A. No bread without Blue Band (margarine) +++

B. No benzene without boiling

C. No base without buffer

D. No bond without breaking

Explanation: It's a lab pun referring to Blue Band margarine.

59. Hydrocarbons are made up of:

A. Carbon and hydrogen only +++

B. Carbon and oxygen

C. Carbon and nitrogen

D. Carbon and sulfur

Explanation: Hydrocarbons consist solely of carbon and hydrogen atoms.

60. The primary source of hydrocarbons is:

A. Coal

B. Petroleum +++

C. Natural gas

D. Wood

Explanation: Petroleum is the primary natural source of hydrocarbons.

61. Which of the following is an aromatic hydrocarbon?

A. Methane

B. Benzene +++

C. Ethane

D. Propane

Explanation: Benzene is a classic aromatic hydrocarbon with a delocalized electron ring.

62. Alkanes are characterized by:

A. Single bonds between carbon atoms +++

B. Double bonds between carbon atoms

C. Triple bonds between carbon atoms

D. Aromatic rings

Explanation: Alkanes contain only single covalent bonds.

63. The general formula for alkanes is:

A.  $C_nH_{2n+2}$  +++

B.  $C_nH_{2n}$

C.  $C_nH_{2n-2}$

D.  $C_nH_n$

Explanation: Alkanes follow the formula  $C_nH_{2n+2}$ .

64. Which of the following is an alkene?

A. Ethane

B. Ethene +++

C. Ethyne

D. Benzene

Explanation: Ethene ( $C_2H_4$ ) contains a C=C double bond, typical of alkenes.

65. The general formula for alkenes is:

A.  $C_nH_{2n+2}$

B.  $C_nH_{2n}$  +++

C.  $C_nH_{2n-2}$

D.  $C_nH_n$

Explanation: Alkenes have formula  $C_nH_{2n}$  due to one double bond.

66. Alkynes are characterized by:

A. Single bonds

B. Double bonds

C. Triple bonds between carbon atoms +++

D. Aromatic rings

Explanation: Alkynes contain at least one carbon-carbon triple bond.

67. The general formula for alkynes is:

- A.  $C_nH_{2n+2}$
- B.  $C_nH_{2n}$
- C.  $C_nH_{2n-2}$  +++
- D.  $C_nH_n$

Explanation: The general formula for alkynes is  $C_nH_{2n-2}$ .

68. Which of the following is a functional group in organic chemistry?

- A. Methyl
- B. Hydroxyl (-OH) +++
- C. Ethyl
- D. Propyl

Explanation: The hydroxyl group (-OH) is a functional group in alcohols.

69. The functional group of aldehydes is:

- A. -COOH
- B. -CHO +++
- C. -OH
- D. -NH<sub>2</sub>

Explanation: The -CHO group is characteristic of aldehydes.

70. The functional group of carboxylic acids is:

- A. -CHO
- B. -COOH +++
- C. -OH
- D. -NH<sub>2</sub>

Explanation: Carboxylic acids contain the -COOH group.



71. Which compound contains an amine group?

- A. Ethanol
- B. Ethylamine +++
- C. Ethanoic acid
- D. Acetone

Explanation: Ethylamine ( $\text{CH}_3\text{CH}_2\text{NH}_2$ ) has the  $-\text{NH}_2$  amine group.

72. Esters are formed by the reaction of:

- A. Alcohol and carboxylic acid +++
- B. Alkene and water
- C. Aldehyde and alcohol
- D. Ketone and amine

Explanation: Esters result from alcohol + carboxylic acid condensation.

73. The functional group of ketones is:

- A.  $-\text{CHO}$
- B.  $-\text{C}=\text{O}$  (carbonyl group) +++
- C.  $-\text{COOH}$
- D.  $-\text{OH}$

Explanation: Ketones contain a carbonyl group bonded to two carbons.

74. Which of the following is a polymer?

- A. Glucose
- B. Polyethylene +++
- C. Ethanol
- D. Benzene

Explanation: Polyethylene is a synthetic polymer made from ethylene.

75. The process of forming polymers from monomers is called:

- A. Hydrolysis
- B. Polymerization +++
- C. Saponification
- D. Esterification

Explanation: Polymerization joins monomers into polymers.

76. Which of the following is a natural polymer?

- A. Polyvinyl chloride
- B. Cellulose +++
- C. Polystyrene
- D. Teflon

Explanation: Cellulose is a natural polymer in plant cell walls.

77. The monomer of proteins is:

- A. Glucose
- B. Amino acid +++
- C. Nucleotide
- D. Fatty acid

Explanation: Proteins are made up of amino acids.

78. The monomer of nucleic acids is:

- A. Amino acid
- B. Nucleotide +++
- C. Glucose
- D. Glycerol

Explanation: DNA and RNA are built from nucleotides.

79. Lipids are primarily composed of:

- A. Amino acids
- B. Fatty acids and glycerol +++
- C. Nucleotides
- D. Monosaccharides

Explanation: Lipids are esters of fatty acids and glycerol.

80. Carbohydrates are made up of:

- A. Amino acids
- B. Monosaccharides +++
- C. Fatty acids
- D. Nucleotides

Explanation: Carbohydrates are composed of monosaccharide units like glucose.

81. The simplest carbohydrate is:

- A. Sucrose
- B. Glucose +++
- C. Starch
- D. Cellulose

Explanation: Glucose is a monosaccharide, the simplest form of carbohydrate.

82. Which of the following is a disaccharide?

- A. Glucose
- B. Sucrose +++
- C. Fructose
- D. Maltose

Explanation: Sucrose is a disaccharide made of glucose and fructose.

83. Polysaccharides include:

- A. Glucose
- B. Starch +++
- C. Sucrose
- D. Fructose

Explanation: Starch is a polysaccharide built from many glucose units.

84. The primary source of energy for the human body is:

- A. Proteins
- B. Carbohydrates +++
- C. Lipids
- D. Nucleic acids

Explanation: Carbohydrates are the main source of quick energy.

85. Which process converts sugars into alcohol?

- A. Fermentation +++
- B. Photosynthesis
- C. Hydrolysis
- D. Oxidation

Explanation: Fermentation is the anaerobic breakdown of sugars into alcohol and CO<sub>2</sub>.

86. The enzyme that breaks down starch is:

- A. Amylase +++
- B. Lipase
- C. Protease
- D. Nuclease

Explanation: Amylase digests starch into simpler sugars.

87. The pH of a neutral solution is:

- A. 7 +++
- B. 4
- C. 9
- D. 14

Explanation: A neutral solution has a pH of 7.

88. An acid is a substance that:

- A. Donates protons ( $H^+$ ) +++
- B. Accepts protons
- C. Releases  $OH^-$
- D. Increases pH

Explanation: Acids are proton donors.

89. A base is a substance that:

- A. Donates protons
- B. Accepts protons ( $H^+$ ) +++
- C. Releases  $H^+$
- D. Decreases pH

Explanation: Bases accept protons and raise pH.

90. The pH of a strong acid solution is typically:

- A. Less than 7 +++
- B. 7
- C. Greater than 7
- D. 14

Explanation: Strong acids have pH values below 7.

91. The pH of a strong base solution is typically:

- A. Less than 7
- B. 7
- C. Greater than 7 +++
- D. 0

Explanation: Strong bases have pH values above 7.

92. Which indicator changes color in acidic conditions?

- A. Phenolphthalein
- B. Litmus (red in acid) +++
- C. Methyl orange
- D. Bromothymol blue

Explanation: Litmus turns red in acidic solutions.

93. Which indicator changes color in basic conditions?

- A. Litmus (blue in base) +++
- B. Methyl orange
- C. Phenol red
- D. Bromocresol green

Explanation: Litmus turns blue in basic solutions.

94. The endpoint of a titration is detected using:

- A. pH meter
- B. Indicator +++
- C. Thermometer
- D. Barometer

Explanation: Indicators show the titration endpoint by changing color.

95. The reaction between an acid and a base is called:

- A. Neutralization +++
- B. Oxidation
- C. Reduction
- D. Sublimation

Explanation: Neutralization produces salt and water.

96. The product of a neutralization reaction includes:

- A. Water and salt +++
- B. Carbon dioxide
- C. Oxygen
- D. Hydrogen gas

Explanation: Neutralization forms water and salt.

97. Which gas is produced when an acid reacts with a metal?

- A. Carbon dioxide
- B. Hydrogen +++
- C. Oxygen
- D. Nitrogen

Explanation: Acid + metal  $\rightarrow$  salt + hydrogen gas.

98. The reaction of a carbonate with an acid produces:

- A. Hydrogen
- B. Carbon dioxide +++
- C. Oxygen
- D. Nitrogen

Explanation: Carbonates release  $\text{CO}_2$  when reacting with acids.

99. The strongest acid among the following is:

- A. Acetic acid
- B. Hydrochloric acid (HCl) +++
- C. Citric acid
- D. Carbonic acid

Explanation: HCl is a strong acid that ionizes completely.

100. The strongest base among the following is:

- A. Ammonia
- B. Sodium hydroxide (NaOH) +++
- C. Calcium hydroxide
- D. Magnesium hydroxide

Explanation: NaOH is a strong base that dissociates completely in water.

**BEST OF LUCK - BIG BOSS**



**Paper 7 of 10 — BAMBOY 100L 2nd semCHE 104  
(BAMBOY)DETERMINATION OF PURITY - CHEM 104- SNOW  
DON.pdf**

Lecture/Notes · Google Drive harvest

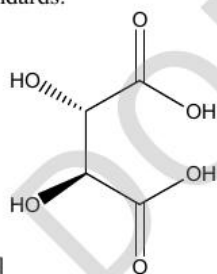
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**EXPERIMENT 1 - Determination of the purity and identity of organic compounds by melting point and/or analytical thin layer chromatography****PART A Melting points and mixed melting points.**

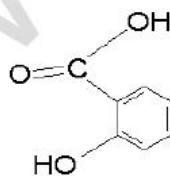
As discussed in the introductory handout, a sharp melting point is generally a reliable indicator of the purity of an organic compound. Solid mixtures tend to show broad melting point ranges. The identity or non-identity of 2 solids, which have similar melting points, can easily be established by carrying out a mixed melting point experiment. Yani, if the melting point does not change upon mixing, then the 2 solids are identical. If the 2 samples have different compositions, a significant lowering and broadening of the melting temperature range will be observed.

**The Experiment.**

You will be supplied with two standards:



- **A:** L-tartaric acid [m.p. 160°C]



- **B:** Salicylic acid [157-159°C]

You will also be supplied with an **unlabeled** sample **C**, that is one of the above two compounds.

Clearly, the melting points of the standard compounds are similar. Thus, a simple melting point determination of the unknown sample C will not allow you to conclusively distinguish whether it is of the type A or the type B.

To determine whether the unknown sample C is A or B, perform the following steps:

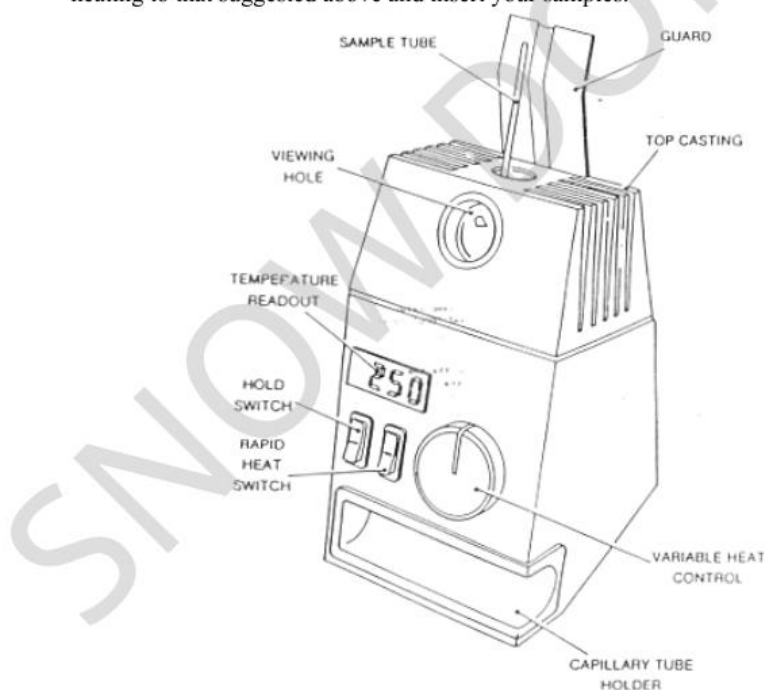
- Determine the melting points of samples A, B and C.
- Grind together a small amount (several mg at most) of your unknown sample C with an equal amount of sample A or sample B to afford two binary mixtures. Determine the melting points of the two mixtures. Draw the appropriate conclusion regarding the identity of your unknown compound. Perform the 3 melting point determinations at the same time.

**Melting point apparatus.**

Please share the two melting point apparatus that are provided. Take turns using the instruments. A schematic drawing of the apparatus and a brief set of instructions may assist your analysis (see below). Please familiarize yourself with the operation of this instrument and if you have any doubts, please consult your demonstrator.

**DETERMINATION OF A MELTING POINT**

Place a few mg of solid material onto a clean surface and grind it to a fine powder with a clean spatula. Note that grinding will not affect the melting point of the sample unless impurities are introduced. The grinding makes it easier to introduce the sample into the capillary tubes. Insert the sample into a capillary tube and pack it firmly by tapping the capillary on the bench. The height of the sample in the capillary should be about 2-3 mm. Insert the sample in the apparatus and determine the melting point. The speed of heating should be adjusted so that near the melting point the rate does not exceed 1-2 degrees per minute. Be patient! To save time, heat the melting apparatus quickly to 80-90°C by setting the heating rate accordingly. Then reduce the rate of heating to that suggested above and insert your samples.

**Basic Procedure:**

1. Load sample crystals into a capillary tube to a depth of about 2mm and tap the bottom of the tube on the bench a fixed number of times. Try tapping five times, for example. The sample depth and packing can affect the result and should be kept as consistent as possible.
2. Insert the sample tubes(s) into the top of the heating block and insert empty sample tubes into any holes not being used.

3. If the approximate melting point of the sample is not known, a quick initial result can be obtained by heating the block rapidly. Yani, turn the control to medium-high initially and note the temperature range at which the sample melts. Remember that at high heating rates, a temperature difference will exist between the thermometer and the sample of up to 15°C. If you were to take the reading under these circumstances, you would be in error, as the apparent melting temperature would be artificially elevated compared to the true value. To obtain a proper reading, please repeat the experiment using a proper temperature gradient.

### CHROMATOGRAPHY BACKGROUND

Chromatography is a term applied to any separation technique that is based on the principle of distributing a sample between a stationary phase and a mobile phase. Simply put, a compound will partition between the mobile phase and stationary phase. In other words, there are moments when a compound is adsorbed/immobilized to the stationary phase and other moments when the same molecule is desorbed/dissolved in a mobile phase. Since the mobile phase is made to flow along a stationary phase, any solutes (eg., the compound) physically move with respect to the stationary phase. Thus, a compound delivered to one end of a chromatography column will adsorb-desorb from the stationary phase and eventually emerge at the other end of the column under the influence of the moving mobile phase. The relative amount of time spent adsorbed in comparison to dissolved is a characteristic function of the compound's partition coefficient under the given experimental conditions. It follows that a separation of two or more components can be achieved whenever their partition coefficients differ.

If one component of the mixture interacts favorably with the stationary phase, a high percentage of molecules of that component will be held immobile. A second component, held weakly, will display a high percentage of molecules in the mobile phase. On average, the molecules of the component that are held less strongly will chromatography (move over the stationary phase in the direction of the flow) at a higher rate than the molecules of the tightly adsorbing compound. In this way, different components will define separate regions or bands along the stationary phase. This separation of the bands is related linearly to the distance travelled on the stationary phase. In other words, easy separations require only a short stationary phase and difficult separations, in which components have very similar distribution coefficients, require a long stationary phase.

#### Types of Chromatography

| <b>Stationary Phase</b>                                                                       | <b>Mobile Phase</b>                                |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------|
| a) <i>Vapour Phase Chromatography (gas-liquid)</i><br>Liquid film adsorbed on a solid support | Helium or nitrogen gas                             |
| b) <i>Column Chromatography (solid-liquid)</i><br>Silica gel, alumina                         | Liquid eg. hexanes, ether, ethyl acetate           |
| c) <i>Thin Layer Chromatography (solid-liquid)</i><br>Silica gel, alumina                     | Liquid eg. hexanes, ether, ethyl acetate, methanol |

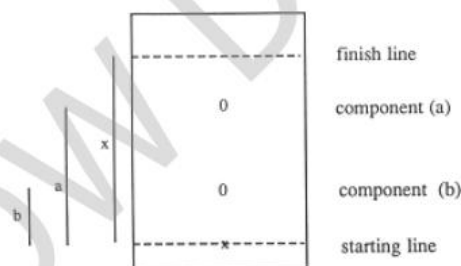
### PART B - Thin Layer Chromatography

In thin layer chromatography, the solvent (mobile phase) will move up a plate coated with silica gel (stationary phase) under the influence of capillary action. The plate is allowed to develop until the solvent has moved a desired distance (finish line). The plate is then removed from the developing chamber and the solvent on the plate is allowed to dry. By this method, components of a mixture are spatially separated.

The material separated on a tlc plate can be visualized in one of several ways:

- Place the plate into an iodine chamber (The iodine usually stains the plate a brown color wherever there is a concentration of organic material).
- Use ultraviolet light (Commercial plates generally contain a fluorescence indicator. Wherever the concentration of a component is sufficient, the fluorescence is extinguished and a spot is seen).
- Char the surface with sulfuric acid or molybdic acid preparations. These reveal dark spots where organic components are concentrated.

The ratio of the rate of migration of a particular component compared to that of the solvent is defined as its  $R_f$  value for that system. In the above diagram the  $R_f$  for component "a" is  $a/x$  and for component "b" is  $b/x$ . The  $R_f$  for a particular substance varies with its structure, the nature of the developing solvent and the type of adsorbant used.



The typical adsorbant of a tlc plate is silica. The surface of the tlc plate is polar because silica is composed of  $\text{-Si-O-Si-O-}$  and  $\text{-Si-OH}$  bonds. Since like prefers like, a natural conclusion is that organic molecules containing polar groups will be adsorbed more strongly than organic molecules not containing polar groups. This adsorption process is mainly governed via dipole-dipole interactions and hydrogen bond interactions. Polar molecules, especially those capable of forming hydrogen bonds with the surface, will migrate more slowly than non-polar molecules (assuming that an appropriate developing solvent has been chosen).

Commonly used developing solvents, in increasing order of polarity, include:

- (1) saturated hydrocarbons (hexanes, petroleum ether)
- (2) aromatic hydrocarbons (toluene)
- (3) halogenated compounds ( $\text{CH}_2\text{Cl}_2$ ,  $\text{CHCl}_3$ )
- (4) ethers, usually diethyl ether
- (5) esters, usually ethyl acetate ( $\text{CH}_3\text{CO}_2\text{C}_2\text{H}_5$ )
- (6) alcohols such as methanol, ethanol

The first goal is to choose a developing system that can separate your components. If the system used is too polar, all the components of the mixture will move very fast ( $R_f$

= 0.9 - 1.0). If the developing solvent is very non-polar and the components are polar, then the  $R_f$  values of the components will be very small ( $<0.1$ ) and again no satisfactory separation will be achieved. Often it is necessary to test many solvent systems in order to realize the one that gives a satisfactory separation. A good starting point for many typical organic mixtures is a solvent system composed of about 2 parts hexanes (a mixture of C-6 hydrocarbons) and 1 part ethyl acetate (relatively polar). Alcohols, which are hydrogen bond acceptors and hydrogen bond donors, are typically more strongly adsorbed onto silica than are comparably sized ketones or ethers which can only accept hydrogen bonds. Hydrocarbons, yani, compounds which contain only carbon and hydrogen, are quite non-polar and tend to have large  $R_f$  values even when non-polar solvents are used as developing solvents.

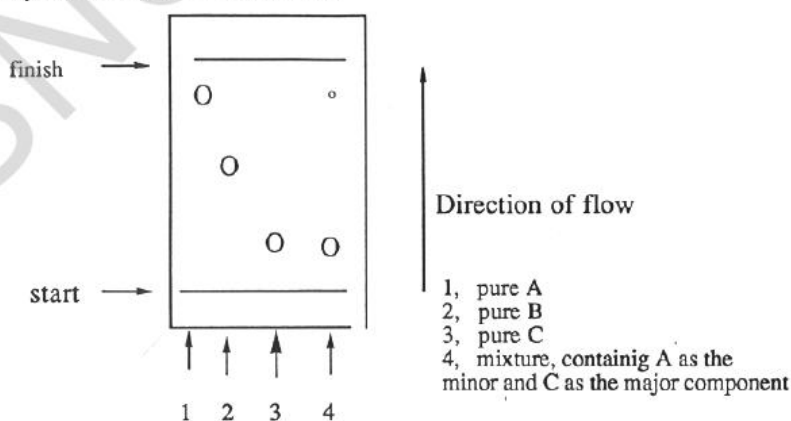
**POLAR**  
 $R'RCHOH$  (alcohols)

**LESS POLAR**  
 $R''RC=O$  (ketones)  
 $R'RCH-O-CH_3$  (ether)

**LEAST POLAR**  
 Hydrocarbons

Based on the discussion above, a pure substance when spotted onto a silica-gel plate and properly developed and visualized, will yield a single spot on the chromatogram. The  $R_f$  value of that spot will vary depending on the polarity of the developing solvent. Similarly, a mixture of two components, when spotted and developed, should afford two spots after development and visualization of the chromatogram. The  $R_f$  values noted will vary depending on the structure of the two components and the choice of the developing solvent. **Conclusion: Only by properly choosing the developing solvent/mixture can two components of an analyte be seen as two distinct spots.**

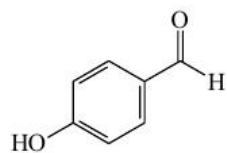
To analyze and identify individual components of a mixture, you should acquire all possible components in pure form (these will serve as standard markers). The analysis is carried out as follows: A small drop of a dilute solution of the mixture and a small drop of each pure component is placed in a separate lane on the silica gel plate. The plate is then developed with an appropriate solvent and the results are visualized. Ideally, the results will be as follows:



### The Experiment

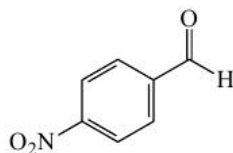
#### The Problem.

You will be supplied with a mixture which may contain a combination of compounds I and II, I and III, or II and III. You also have pure samples of I, II and III. Your goal is to determine the components of your mixture using analytical tlc.



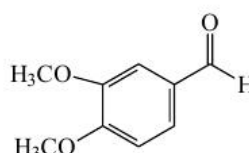
I

4-hydroxybenzaldehyde



II

4-nitrobenzaldehyde



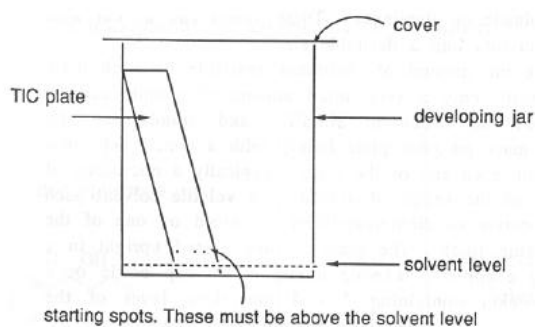
III

3,4-dimethoxybenzaldehyde

Thin layer plates suitable for analytical techniques can be prepared or purchased. Tlc plates are commercially available in 20x20cm sheets consisting of a 0.1mm layer of adsorbant bound to a 0.2mm thick sheet of plastic or aluminum metal. The sheet can be cut with ordinary scissors into a desirable size. Prepare solutions by dissolving about 5mg (several crystals) of I, II, III, and your unknown mixture. Use approximately 1ml of dichloromethane in four separate test tubes or Eppendorf tubes. Using the micro-pipette tips supplied by the demonstrator, spot each solution carefully onto a tlc plate (described below). Check the results with the ultraviolet lamp (use 254nm) to make certain that a small dark spot is visible at each starting point. Do not use too much material! If the spot is too large, the plate will be overloaded and separation may be poor. It may be useful to practice your spotting technique on a strip of 'waste' TLC plate.

In order to simplify and standardize the spotting/developing procedure, lightly mark on your plate with a pencil two lines 0.8cm from each end of the plate. Typically, a microdrop of a solution of the sample dissolved in a volatile solvent such as ethyl acetate or dichloromethane is placed on one of the lines (starting line). The plate is then placed upright in a developing chamber - a small jar or a covered beaker. The vessel should contain a very shallow layer (eg., 3ml total volume) of the developing solvent and its walls can be lined with some filter paper.

Try a 4:1 mixture of hexane/ethyl acetate as an initial condition. You can see a typical set-up below.



Allow the solvent to migrate to the pre-determined point [marked with a pencil] near the top of the plate. Remove the plate and allow it to dry. Visualize the results with a UV lamp. Record the results in your notebook by drawing a sketch of the TLC plate. Determine the  $R_f$  values of each of the three pure compounds and those of the components of your mixture (please use a ruler!). Discuss the composition of your mixture in light of your results.

Repeat the above procedure but develop the plate in pure hexane. Run a third chromatogram using a 1:1 mixture of hexane and ethyl acetate. Again record the visual results and determine the  $R_f$  values of the components. Each partner should tape at least one TLC plate in his/her notebook as part of their experimental results.

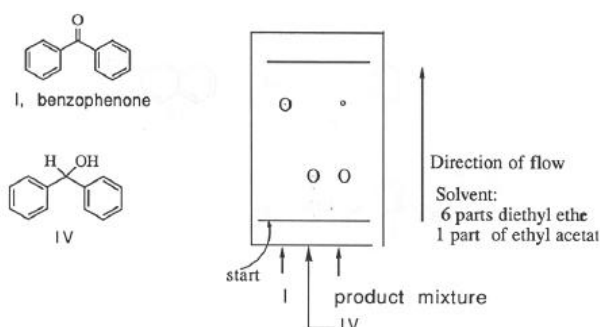
Give a reasonable explanation for the relative  $R_f$  values for I, II, and III. Hint – think of the probable interaction of each compound with silica. If the  $R_f$  values varied in different developing solvents, try to explain why. Which of the three developing solvents were most useful in allowing you to determine the composition of your mixture?

### Questions

1. You have synthesized a compound, which you suspect is aspirin. One evidence in particular was the observed melting point of 133-135°C which is near the literature melting point of aspirin, 135°C. You have access to an authentic sample of aspirin. Describe, in less than three lines, a method based on the melting point technique which would enable you to determine whether your suspicion is true.

2. Benzophenone (I) can be reduced to afford benzhydrol (IV). Ibrahim Talises carried out the experiment and analyzed the starting material and product using TLC. His results are shown below. The TLC shown was obtained using 6 parts of diethyl ether and 1 part of ethyl acetate as developer.

- What conclusion can be drawn about the purity of the product?
- What is the  $R_f$  of benzophenone under these conditions?
- What is the  $R_f$  of benzhydrol?
- Give an explanation of the qualitative difference in the  $R_f$  of the two compounds.
- When the solvent is changed to 1 part ethyl ether and 1 part ethyl acetate, the  $R_f$  of
  - benzhydrol is expected to be a) smaller, b) the same, c) larger?
  - benzophenone is expected to be a) smaller, b) the same, c) larger?





**Paper 8 of 10 — BAMBOY 100L 2nd semCHE 104  
(BAMBOY)IDENTIFICATION OF ORGANIC COMPOUND- SNOW  
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## Identification of Organic Compound by Organic Qualitative Analysis

A Systematic Scheme for the identification of the organic compound is outlined below.

1. Preliminary Test
2. Solubility Test
3. Detection of Extra Elements
4. Detection of Functional Group
5. Determination of physical constant and Identification of the compound.
6. Confirmatory Test.
7. Derivatives preparation and its m.pt determination .

### 1. Preliminary Test

| Test                                                                                                                                        | Observation                                        | Inferences                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| (a) Nature                                                                                                                                  | i) Solid                                           | Carbohydrate, acid, phenol, amine, higher hydrocarbon may be present.                                              |
|                                                                                                                                             | ii) Liquid                                         | Alcohol, ketone, aldehyde, ester, phenol, amines may be present.                                                   |
| (b) Colour                                                                                                                                  | i) Yellow -Solid                                   | m- Dinitrobenzene, p- Nitro toluene, nitro phenol, nitro aniline.                                                  |
|                                                                                                                                             | ii) Yellow -liquid                                 | Nitrobenzene.                                                                                                      |
|                                                                                                                                             | iii) Brown                                         | P – Toluidine, resorcinol.                                                                                         |
|                                                                                                                                             | iv) Blackish                                       | $\alpha$ – Naphthol                                                                                                |
|                                                                                                                                             | v) Pink                                            | $\beta$ – Naphthol                                                                                                 |
|                                                                                                                                             | vi) Buff or reddish                                | Aniline, phenol, Aromatic amine.                                                                                   |
|                                                                                                                                             | vii) Colourless                                    | Simple acid, alcohol, ester, ketone aromatic hydrocarbon.                                                          |
| (c) Odour                                                                                                                                   | i) Carbolic                                        | Phenol, cresol.                                                                                                    |
|                                                                                                                                             | ii) Fishy                                          | Amine                                                                                                              |
|                                                                                                                                             | iii) Sweet pleasant                                | Ester, alcohol and halogen derivatives.                                                                            |
|                                                                                                                                             | iv) Bitter almonds                                 | Nitrobenzene, Benzaldehyde                                                                                         |
|                                                                                                                                             | v) Moth balls                                      | Naphthalene                                                                                                        |
|                                                                                                                                             | vi) No particular smell                            | Aromatic acid, amide, carbohydrate.                                                                                |
| (d) Flame Test                                                                                                                              | i) Sooty flame                                     | Aromatic compound or aliphatic compound containing small proportion of hydrogen e.g. $\text{CHCl}_3, \text{CCl}_4$ |
|                                                                                                                                             | ii) Non sooty flame                                | Aliphatic compound                                                                                                 |
|                                                                                                                                             | iii) Substance chars                               | Carbohydrate, sulphanilic acid.                                                                                    |
| (f) Test for unsaturation<br>1. $\text{KMnO}_4$ test.<br>Substance + 2 ml of water shake well + 2 drops of dilute $\text{KMnO}_4$ solution. | i) Decolourisation of $\text{KMnO}_4$              | Unsaturated or easily oxidizable compound.                                                                         |
|                                                                                                                                             | ii) No decolourisation                             | Saturated compound.                                                                                                |
| 2. Bromine water Test.<br>Substance + 2 ml of water shake well + 2 drops of bromine water.                                                  | i) Decolourisation of bromine water.               | Unsaturated compound.                                                                                              |
|                                                                                                                                             | ii) Decolourisation with formation of precipitate. | Easily substituted compound.                                                                                       |
|                                                                                                                                             | iii) No decolourisation.                           | Saturated compound.                                                                                                |

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## 2. Solubility Test

| <b>(A) Perform the following test only if the substance is soluble /miscible in water</b>                                                                                             |                                               |                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------|
| <b>Test</b>                                                                                                                                                                           | <b>Observation</b>                            | <b>Inferences</b>                                    |
| (a). 0.1gm of substance + 3ml of water shake well.<br><b>Test the solution with litmus paper.</b>                                                                                     | Substance dissolves                           | Lower member of alcohol, ester ketone, carbohydrate. |
|                                                                                                                                                                                       | i) <b>Blue</b> litmus paper turns <b>red</b>  | Water soluble acid or phenol present.                |
|                                                                                                                                                                                       | ii) <b>Red</b> litmus paper turns <b>blue</b> | Water soluble base present                           |
|                                                                                                                                                                                       | iii) No action on the litmus paper            | Water soluble neutral present                        |
| (b). 0.1gm of substance + Saturated $\text{NaHCO}_3$ solution. Strong effervescence and substance dissolves.<br><br>To this clear solution add <b>conc. HCl</b>                       | <b>No solid appear</b>                        | Water soluble acid present                           |
| (c). 0.1gm of substance + water shake well, substance dissolves.<br><br>To this clear solution add alcoholic $\text{FeCl}_3$ Solution.                                                | Blue to violet colour                         | Water soluble phenol present                         |
| <b>(B) Perform the following test only if the substance is insoluble / immiscible in water</b>                                                                                        |                                               |                                                      |
| (a) 0.1gm of substance +3 ml Saturated $\text{NaHCO}_3$ solution. Shake well. The substance dissolves.<br><b>To this clear solution add conc. HCl drop by drop.</b>                   | Strong effervescence                          | Carboxylic acid present.                             |
|                                                                                                                                                                                       | A solid appear                                | Carboxylic acid confirmed.                           |
| (b) 0.1gm of substance +3 ml dilute $\text{NaOH}$ solution.,Shake well. The substance dissolves.<br><b>To this clear solution add conc. HCl drop by drop.</b>                         | A solid or emulsion appear                    | Phenol present                                       |
| (c) 0.1gm of substance +3 ml 1:1 $\text{HCl}$ solution. Shake well. The substance dissolves.<br><b>To this clear solution add 20% <math>\text{NaOH}</math> solution drop by drop.</b> | A solid appear                                | Base confirmed                                       |
| (d) If substance is insoluble in $\text{NaHCO}_3, \text{NaOH}, \text{HCl}$ solution.                                                                                                  | .....                                         | Neutral compound present                             |

## 3. Detection of Extra Elements

### Sodium Fusion Test (Lassaigne's Test)

1. Take a small piece of dry sodium metal in a fusion tube and heat it gently till the metal melts or fuses.
2. Add equal quantity of compound to this fused metal [ If the compound is a liquid then add two drops of it with a capillary]
3. Heat it gently then strongly till it becomes red hot.
4. Plung the red hot tube in 10 ml or  $\frac{3}{4}$  of a test tube of distilled water taken in a porcelain dish, covering it immediately with an asbestos sheet crush the fusion tube completely.
5. Carry out more fusion in the similar way. Boil the extract for five minutes reduce the volume to about 5 ml and filter. Perform following test using this filtrate.

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### Detection of Elements

| Test                                                                                                                                                                                                                               | Observation                                                | Inferences                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------|
| <b>Test for Nitrogen</b>                                                                                                                                                                                                           |                                                            |                                          |
| 1 ml of extract + 2-3 drops of NaOH solution to make it alkaline + a few drops of freshly prepared $\text{FeSO}_4$ solution, boil for a few minutes, cool and acidify it with by adding dil. HCl or dil. $\text{H}_2\text{SO}_4$ . | Blue or green colour solution or Prussian blue coloration. | <b>Nitrogen present.</b>                 |
| <b>Test for Sulphur</b>                                                                                                                                                                                                            |                                                            |                                          |
| (i) 1 ml of extract + 1 ml of 2N Acetic Acid + 1 ml of Lead Acetate Solution.                                                                                                                                                      | A black precipitate.                                       | <b>Sulphur present.</b>                  |
| (ii) 1 ml of extract + 1 drop of sodium nitroprusside solution.                                                                                                                                                                    | A violet or purple coloration.                             | <b>Sulphur present.</b>                  |
| (iii) 1 ml of extract + 1 ml of aqueous $\text{FeCl}_3$ solution.                                                                                                                                                                  | A blood red coloration                                     | <b>Nitrogen and Sulphur are present.</b> |
| <b>Test for Halogen</b>                                                                                                                                                                                                            |                                                            |                                          |
| (i) 1 ml of extract + 1 ml of dilute $\text{HNO}_3$ (boil well if N and S are present) + 1 ml of 5% $\text{AgNO}_3$ solution.                                                                                                      | A thick white precipitate.                                 | <b>Halogen present.</b>                  |
| (ii) If halogen is present carry out the following test:<br>1ml of extract + 1ml of dilute $\text{H}_2\text{SO}_4$ + 0.5ml of $\text{CHCl}_3$ and 0.5ml of chlorine water, shake well and observe the colour of chloroform layer.  | (i) Violet colour                                          | Iodine present.                          |
|                                                                                                                                                                                                                                    | (ii) Yellow or brown colour.                               | Bromine present.                         |
|                                                                                                                                                                                                                                    | (iii) Colourless layer                                     | Chlorine present.                        |

### Classify the given compound on the Basis of Element Present

On the basis of the elements present in the organic compound, it belongs to one of the four groups, which may be further divided in subgroups as follows :

|                                                                          |                                         |
|--------------------------------------------------------------------------|-----------------------------------------|
| <b>Group I : C, H, (O)</b>                                               | <b>Group III : C, H, (O), N and S</b>   |
| (i). Carboxylic acids<br>(ii). Phenol<br>(iii). Neutrals                 | (i). Acids<br>(ii). Neutrals            |
| <b>Group II : C, H, (O) and N</b>                                        | <b>Group IV : C, H, (O) and Halogen</b> |
| (i). Carboxylic acids<br>(ii). Phenols<br>(iii). Bases<br>(iv). Neutrals | (i). Neutrals                           |

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#### 4. Detection of Functional Groups

| Group I : C, H, (O) Carboxylic acids                                                                                                                                                                                                         |                                                                 |                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| Test                                                                                                                                                                                                                                         | Observation                                                     | Inferences                                                         |
| (a) 0.1gm of substance +3 ml Saturated $\text{NaHCO}_3$ solution. Shake well. The substance dissolves.<br><b>To this clear solution add conc. HCl drop by drop.</b>                                                                          | Strong effervescence                                            | Carboxylic acid present.                                           |
|                                                                                                                                                                                                                                              | A solid appear                                                  | Carboxylic acid confirmed.<br>[ F.G $\rightarrow$ $-\text{COOH}$ ] |
| (b) 0.05 gm of compound + 1ml of water , shake well + 1-2 drops of alcoholic $\text{FeCl}_3$ solution.                                                                                                                                       | i. Buff coloured precipitate                                    | Benzoic acid or phthalic acid.                                     |
|                                                                                                                                                                                                                                              | ii. Violet coloured precipitate                                 | Salicylic acid                                                     |
|                                                                                                                                                                                                                                              | iii. Violet coloured precipitate obtain on heating the solution | Acetyl salicylic acid                                              |
|                                                                                                                                                                                                                                              | iv. Yellow coloured precipitate                                 | Cinnamic acid                                                      |
|                                                                                                                                                                                                                                              | v. Faint reddish coloured precipitate                           | Succinic acid                                                      |
|                                                                                                                                                                                                                                              | vi. Deep yellow coloured solution                               | Citric acid                                                        |
|                                                                                                                                                                                                                                              | vii. No change in $\text{FeCl}_3$ solution.                     | Oxalic acid                                                        |
| Group I : C, H, (O) Phenol                                                                                                                                                                                                                   |                                                                 |                                                                    |
| (a) 0.1gm of substance +3 ml dilute NaOH solution. Shake well.<br><b>To this clear solution add conc. HCl drop by drop.</b>                                                                                                                  | Compound dissolves completely.                                  | Phenol present                                                     |
|                                                                                                                                                                                                                                              | A solid or emulsion appear                                      |                                                                    |
| (b).0.01gm of compound + 3ml of water or alcohol shake well and add a drop of alcoholic or neutral $\text{FeCl}_3$ solution.                                                                                                                 | i. Violet Colour                                                | <b>Phenol Present</b>                                              |
|                                                                                                                                                                                                                                              | ii. Blue Violet Colour                                          | Resorcinol                                                         |
|                                                                                                                                                                                                                                              | iii . White ppt slowly changing to pink, blue or violet         | $\alpha$ - naphthol                                                |
|                                                                                                                                                                                                                                              | iv. Green Colour                                                | $\beta$ - naphthol                                                 |
| <b>(c). Phthalein test</b><br>0.01gm of compound + 0.01gm of phthalic anhydride + 2 drops of conc. $\text{H}_2\text{SO}_4$ .Heat gently until the mixture fuses. Cool and pour it in a beaker containing 20 ml of very dilute NaOH solution. | Pink colour                                                     | <b>Phenol Present</b>                                              |
|                                                                                                                                                                                                                                              | Green or bluish green                                           | $\alpha$ - naphthol<br>$\beta$ - naphthol                          |
|                                                                                                                                                                                                                                              | Yellowish-green fluorescence                                    | Resorcinol                                                         |
| <b>(d).Liebermann test.</b><br>0.01gm of compound+1ml of conc. $\text{H}_2\text{SO}_4$ + 2 crystal of $\text{NaNO}_2$ . Heat it gently. Dilute it with water , add 20% NaOH solution.                                                        | Red colouration<br>Bluish greenish coloration                   | [ F.G $\rightarrow$ $-\text{OH}$ (Phenolic) ]                      |

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| Group I : C, H, (O) Neutrals                                                                                                                                                                                                                                                                                                        |                                                                       |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Test                                                                                                                                                                                                                                                                                                                                | Observation                                                           | Inferences                                                                                         |
| <b>(a) Test for Carbohydrates</b><br><b>Molish Test</b> : ( Perform this test only if the compound is colorless and soluble in water )<br>Dissolve 0.5 gm of the compound in 2ml of water + 2/3 drops of 10% $\alpha$ -naphthol dissolved in ethyl alcohol, add carefully 1 ml of conc. $H_2SO_4$ along the sides of the test tube. | A Violet ring appears at the junction of two layers.                  | Carbohydrate present                                                                               |
| <b>(b) Test for Aldehydes and Ketones.</b><br>(i) 0.05 gm of the compound + 3 ml of 2,4 dinitrophenyl hydrazine. Shake well.                                                                                                                                                                                                        | Yellow or orange red crystalline precipitate                          | Aldehyde or Ketone Present                                                                         |
| If this test is positive, perform the following test to distinguish between aldehyde and ketone.                                                                                                                                                                                                                                    |                                                                       |                                                                                                    |
| <b>Test for Aldehydes :</b><br><b>(i) Schiff's Test</b><br>0.05 gm of the compound + 2/3 ml of Schiff's Reagent .Shake well.                                                                                                                                                                                                        | Violet colour immediately develops                                    | Aliphatic aldehyde present                                                                         |
|                                                                                                                                                                                                                                                                                                                                     | Pink colour slowly develops                                           | Aromatic aldehyde present                                                                          |
| <b>(ii) Tollen's Test OR Silver Mirror test:</b><br>0.1 gm of the compd +2-3mlTollents reagent (i.e. Ammonical silver Nitrate solution) + Heat it on a boiling water bath.                                                                                                                                                          | A silver mirror is formed on the inner sides of the test tube         | Aldehyde present<br>[ F.G $\rightarrow$ $-CHO$ ]                                                   |
| <b>(iii) Fehling Solution Test</b><br>0.1gm of the compd + 1ml Fehling A + 1ml Fehling B solution .Heat it gently                                                                                                                                                                                                                   | Formation of red ppt of Cuprous oxide                                 |                                                                                                    |
| <b>(iv) Benedict's test</b><br>0.1gm of the compd + Benedicts solution + Heat it gently.                                                                                                                                                                                                                                            | Formation of red ppt<br>Cuprous oxide                                 |                                                                                                    |
| <b>Test for Ketones :</b><br>0.1gm of compd + 2ml of sodium nitroprusside solution + 2 drops of NaOH                                                                                                                                                                                                                                | Wine Red colour or Orange red colour<br>$CH_3-CO-$ gr gives this test | Ketone present<br>[ F.G $\rightarrow$ $\begin{array}{c} \diagup \\ C=O \\ \diagdown \end{array}$ ] |
| <b>(c). Test for Esters</b><br>Dissolve 0.1gm or 0.5ml of compound in 1 ml of ethyl alcohol + a drop of phenolphthalein + 2 drops of very dilute NaOH solution. Heat on a boiling water bath.                                                                                                                                       | Pink colour disappears                                                | Ester present                                                                                      |
| <b>(d). Test for Alcohols</b><br>(i). Take a small piece of dry Na metal in a fusion tube and add a few drops of compound.<br>(ii). 1ml of acetyl chloride in a dry test tube + drops of the compound.                                                                                                                              | Rapid evolution of $H_2$<br><br>Strong effervescence                  | Alcohol present<br><br>Alcohol present                                                             |
| <b>(e). Test for Hydrocarbons</b><br>0.5ml of compound + 1 ml of iodine in carbon disulphide, shake well                                                                                                                                                                                                                            | Purple colour of $CS_2$ layer changes to brown colour                 | Ether present                                                                                      |
| <b>(f). Test for Hydrocarbons</b><br>0.1gm of compound + 1-2 ml of water, shake well + 1-2 drops of very very dilute $KMnO_4$ solution. Shake again.                                                                                                                                                                                | If all the above tests fail                                           | Hydrocarbon present                                                                                |
|                                                                                                                                                                                                                                                                                                                                     | Decolourisation                                                       | Unsaturated hydrocarbon present                                                                    |
|                                                                                                                                                                                                                                                                                                                                     | No decolourisation                                                    | Saturated hydrocarbon present                                                                      |

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| Group II : C, H, (O) and N Carboxylic acids                                                                                                                                                                                                                                 |                                                                                                             |                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Test                                                                                                                                                                                                                                                                        | Observation                                                                                                 | Inferences                                                             |
| i) 0.1 gm of the compd + 2/3 ml of sat NaHCO <sub>3</sub> solution. Shake well .<br>To this clear solution add conc. HCl drop by drop                                                                                                                                       | Strong effervescences and compound dissolves.                                                               | Nitro-Carboxylic acid or amino carboxylic acid present                 |
|                                                                                                                                                                                                                                                                             | Solid appears                                                                                               | Nitro-Carboxylic acid present                                          |
|                                                                                                                                                                                                                                                                             | No solid appears                                                                                            | Amino carboxylic acid present                                          |
| ii) If nitro carboxylic acid is present, perform the following test of nitro group –<br>a) Test for nitro group – 0.2 gm of the compd + 2 ml of ethyl alcohol + 0.1 gm of solid NH <sub>4</sub> Cl + 0.1 gm of Zn dust. Boil for 5min and filter. Filter + Tollen's Reagent | Black or grey precipitate                                                                                   | Nitro group present.<br>[ F.G → – NO <sub>2</sub> ]                    |
| iii) If amino carboxylic acid present ,perform the following test for amino gr-<br>0.5 gm of the compd +3 / 4 ml of 1:1 HCl, shake well. Cool and add few drops of 2% NaNO <sub>2</sub> solution                                                                            | A clear solution is obtained which when added to a cold solution of alkaline β-naphthol give orange red dye | Aromatic primary amino group is present<br>[ F.G → – NH <sub>2</sub> ] |
| Group II : C, H, (O) and N Phenols                                                                                                                                                                                                                                          |                                                                                                             |                                                                        |
| i)0.1 gm of the compd + dilute NaOH solution<br>Add conc HCl drop by drop                                                                                                                                                                                                   | Compd dissolves producing deep yellow or orange colour                                                      | Nitro phenol or Amino phenol is present                                |
|                                                                                                                                                                                                                                                                             | Solid reappears                                                                                             | Nitrophenol present                                                    |
|                                                                                                                                                                                                                                                                             | No solid reappears                                                                                          | Aminophenol present                                                    |
| ii) If nitro phenol present perform the test for –NO <sub>2</sub> group as written above.                                                                                                                                                                                   | Black or grey ppt                                                                                           | Nitro group present                                                    |
| iii) If Aminophenol present, perform the test for –NH <sub>2</sub> , group as written above.                                                                                                                                                                                | Orange red dye                                                                                              | Aromatic Primary amino group present                                   |
| Group II : C, H, (O) and N Bases                                                                                                                                                                                                                                            |                                                                                                             |                                                                        |
| 0.5 gm of the compd +3 / 4 ml of 1:1 HCl, shake and filter. Filtrate + 20% NaOH drop by drop                                                                                                                                                                                | Compd dissolves                                                                                             | Amines present                                                         |
|                                                                                                                                                                                                                                                                             | Solid reappears or emulsion obtained                                                                        | Amines confirmed                                                       |
| 0.5 gm of the compd +3 / 4 ml of 1:1 HCl, shake well. Cool and add few drops of 2% NaNO <sub>2</sub> solution                                                                                                                                                               | Deep yellow, red or green colour                                                                            | Aromatic tertiary amino group (-N-) present                            |
|                                                                                                                                                                                                                                                                             | White or yellow emulsion                                                                                    | Aromatic secondary amino group (-NH-) present                          |
|                                                                                                                                                                                                                                                                             | A clear solution is obtained which when added to a cold solution of alkaline β-naphthol give orange red dye | Aromatic primary amino group (-NH <sub>2</sub> ) present               |

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| Group II : C, H, (O) and N Neutrals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                      |                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Observation                                                                                                                                                          | Inferences                                                                                                                                                                                                                                 |
| <b>(1). Test for Diphenylamine</b><br>1/2 crystal of compd + 1 ml of conc.H <sub>2</sub> SO <sub>4</sub> + Conc HNO <sub>3</sub> . or one drop of dilute NaNO <sub>2</sub> solution (If this test is positive,perform group test listed under C,H,(O) and N Basic to confirm the functional group)                                                                                                                                                                                                                                                                                                                                                                                                                  | Deep Blue Colour                                                                                                                                                     | Diphenylamine present                                                                                                                                                                                                                      |
| <b>(2) Test for Amides</b><br>0.2 gm of compound + 3ml of 20% NaOH sol. Boil for 1-2 min. Test the gas evolved -<br><br>i). Bring a moist red litmus paper over the mouth of test tube<br><br>ii). Dip a glass rod in conc HCl and hold this over the mouth of the test tube                                                                                                                                                                                                                                                                                                                                                                                                                                        | Evolution of NH <sub>3</sub> (confirm with the smell)<br><br>Red litmus paper turns blue<br><br>Dense white fumes of NH <sub>4</sub> Cl are evolved on the glass rod | Amide group present<br><br>Amide group confirmed<br><br>$[ \text{F.G} \rightarrow \text{—}\overset{\text{O}}{\underset{\text{  }}{\text{C}}}\text{—NH}_2 ]$<br>Amide                                                                       |
| <b>(3) Test for Anilide</b><br>0.1 gm of the compd + 1 ml of conc HCl. Boil for 2 min, cool and add 5 ml of water + a few drops of cold NaNO <sub>2</sub> sol. And mix well. Add this solution to a cold solution of alkaline β-naphthol.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Orange red dye                                                                                                                                                       | Anilide Group present<br>$[ \text{F.G} \rightarrow \text{Ar} - \text{NH} - \text{CO} - \text{R} ]$<br>$\text{R} \rightarrow -\text{CH}_3, -\text{C}_6\text{H}_5$                                                                           |
| <b>(4). Test for Nitro group</b><br><b>(a). Mulliken's Test</b><br>0.2 gm of the compd + 2 ml of ethyl alcohol + 0.1 gm of solid NH <sub>4</sub> Cl + 0.1 gm of Zn dust. Boil for 5min and filter. Filter + Tollen's Reagent<br><b>(b). Azo-Dye Test</b><br>(If primary amino group absent, then only perform this test)<br>0.5 gm of the compd + 0.5 gm of Tin metal + 2 ml of conc HCl. Boil or 3 min, cool filter and dilute with about 5 ml of water + few drops of NaNO <sub>2</sub> . Add this sol to a cold solution of alkaline β-naphthol.<br><b>(c). Test for Dinitro compound</b><br>0.5 g of the compound + 1-2 ml of acetone, shake well to dissolve the compound + 1-2 drops of dilute NaOH solution. | Black or grey precipitate<br><br><br><br><br><br><br><br><br><br>Orange red dye<br><br><br><br><br><br><br><br><br><br>Dark purple or Violet colour                  | Nitro group present.<br>$[ \text{F.G} \rightarrow \text{—NO}_2 ]$<br><br><br><br><br><br><br><br><br><br>Nitro group present.<br>$[ \text{F.G} \rightarrow \text{—NO}_2 ]$<br><br><br><br><br><br><br><br><br><br>Dinitro compound present |
| Group III : C, H, (O) and N & S - Acids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                      |                                                                                                                                                                                                                                            |
| This class includes amino sulphonic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                      |                                                                                                                                                                                                                                            |
| <b>(1) Test for Amino Sulphonic Acid</b><br><b>a)</b> 0.2 gm of the compound + 3 -4 ml of saturated NaHCO <sub>3</sub> solution.<br><br><b>b)</b> Perform test for —NH <sub>2</sub> group as above.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Effervescences and compound dissolves<br><br><br>Orange red dye is formed                                                                                            | Amino Sulphonic acid present<br><br><br>Aromatic primary amino group present                                                                                                                                                               |



Dr.Mohammad Idrees and Dr.Naqi.J.Siddiqui, Assistant Professor, Department of Chemistry, Institute of Science, Nagpur

| Group III : C, H, (O) and N & S - Neutral                                                                                                                                                   |                                            |                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| This class includes Thiourea                                                                                                                                                                |                                            |                                                                                                                                                                   |
| <b>Test for Thiourea</b><br>Thiourea is soluble in water and neutral to litmus paper<br>0.1 gm of comp + 2ml of 20% NaOH solution, boil, cool and add a few drops of lead acetate solution. | Black ppt is obtained.                     | <b>Thiourea is confirmed</b><br>$\text{[ F.G } \rightarrow \text{ } \begin{array}{c} \text{S} \\    \\ \text{---C---NH}_2 \end{array} \text{ ]}$ <b>Thioamide</b> |
| Group IV : C, H, (O) and Halogen - Neutral                                                                                                                                                  |                                            |                                                                                                                                                                   |
| 0.2 gm of compd + 2-3 of ml of dilute NaOH solution Boil for a few min, cool + dilute HNO <sub>3</sub> till acidic + 1ml of AgNO <sub>3</sub> solution.                                     | White or yellow ppt<br><br>No Precipitate. | Aliphatic halide like CHCl <sub>3</sub> , CCl <sub>4</sub> present<br>Chlorobenzene or bromobenzene present                                                       |

### 5. Determination of physical constant and Identification of the compound.

Take organic compound into one end sealed capillary and determine m.pt /B pt

### 6. Confirmatory Test

| Test | Observation | Inference |
|------|-------------|-----------|
|      |             |           |

### 7. Derivatives preparation and its m.pt determination .

Results.

From the above step by step analysis of given organic compound shows that, it is

State :  
 Colour :  
 Odour :  
 Aromatic / Aliphatic :  
 Saturated /unsaturated :  
 Solubility :  
 Extra elements :  
 Functional group :  
 M.pt/B . pt

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**Paper 9 of 10 — BAMBOY 100L 2nd semCHE 104  
(BAMBOY)RAMSEY CHE 104 possible questions the blueprint  
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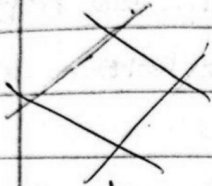
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CHE 104 TUTORIAL

- Try the Past questions you see and those with diagrams.
- Wash bottle, test tube,



They can bring a table  
test for saturated, unsaturated,  
solubility test

- A diagram; what is this set-up:



Filtration set-up.

- Boiling point determination.
  - We can use boiling point for several tests.
  - The separation of liquids with different boiling points Distillation.
  - Close <sup>B.P</sup> - Fractional distillation
  - wide difference - simple distillation.
  - Organic solvent have B.P <sup>lower</sup> than that of water. → E.g of organic solvent -
  - The B.P is lower because;
  - Because of the <sup>weak</sup> intermolecular force between them
- B.P — Boiling Point.

Beloved $t^\circ$  - temperature.

e.g. Vanderwaals force.

- Internal & External ~~free~~ Pressure must be equal before boiling takes place.
- We use a thermometer to measure the B.P of a
- The thermometer ~~be~~ <sup>should</sup> not touch the bottom of the flask.

\* Functions of checking for Boiling Points.

- To identify compounds
- To check for purity
- To separate mixtures
- The presence of impurities increases the boiling point of water.
- Boiling commences when the <sup>External</sup> Atmospheric and Vapour Pressure (Internal) are equal.
- At s.t.p  $H_2O$  boils at  $100^\circ C$
- The Pressure affects the  $t^\circ$  of a solution.

\* Examples of Compds and their boiling points:-

|                                                   |      |   |                                          |        |
|---------------------------------------------------|------|---|------------------------------------------|--------|
| CHCl <sub>3</sub><br>Chloroform                   | 61°C | ↑ | C <sub>6</sub> H <sub>6</sub><br>Benzene | - 80°C |
| (normal) n-Heptane C <sub>7</sub> H <sub>16</sub> | 98°C |   | Ethanol C <sub>2</sub> H <sub>5</sub> OH | - 78°C |

$C_6H_{14}$   
n-Hexane -  $69^\circ C$

$CH_3-CO-CH_3$   
Propan-2-one  
 $C_3H_6O$

Acetone - ~~B.P.~~  $56^\circ C$

- The higher the molecular weight, the higher the boiling point.

Chloroform  $CHCl_3$

Benzene B.P. > n-Hexane B.P. Because it has 3 conjugate double bonds.

- Hydrogen bonds are stronger than Vander waals forces (bond). The presence of ~~ben~~ hydrogen bond increase B.P.
- The higher the altitude the lower the boiling point.

- Why do the presence of impurities increase boiling point? because the impurities limit or lower the vapour pressure of the liquid.

- Mention 2 forces that affect the B.P of a substance.
  - Hydrogen bond  $\downarrow$  Strong Higher B.P
  - Vander waal forces; dipole,  $\downarrow$  Weak Lower B.P

### Distillation

- We use distilled water.

RELOVED.

BELOVED

- Separating liquids of different B.P. - Distillation
- For Immiscible liquid  $\rightarrow$  steam distillation (and natural product)
- For High B.P substances that are heat sensitive - Vacuum distillation.

- ① Organic solvent has generally Lower B.P compared to water
- ② Ether boils at a lower  $t^\circ$  because of Weak Intermolecular forces like Van der Waals.
- ③ If the atmospheric pressure decreases, the boiling point Decreases.
- ④ A mixture of two immiscible liquids usually boil at different temperatures. (Lower / higher than one another)
- ⑤ The boiling point of chloroform is  $61^\circ\text{C}$ .
- ⑥ The instrument used to measure B.P is a Thermometer.
- ⑦ Impurities in liquids increases B.P.
- ⑧ A liquid boils when the external pressure is equal to the internal pressure.
- ⑨ If pressure affects B.P, the B.P of a particular substance (solution) <sup>may</sup> be different at different location because of different Atmospheric pressures.
- ⑩ The why will ether boil at a lower temp. than water. because of weak Intermolecular forces.

⑪ How do you place the thermometer in the soln \_\_\_\_\_  
inside the solution without touching the bottom of  
the flask.

⑫ In crude oil separation, we use fractional distillation

⑬ What do we use to measure the density of a solvent?  
Hychrometer or Pycnometer.

⑭ ~~To determine the~~

⑮ The melting point is the best way of determining  
the purity of a substance. pure substances have  
sharp melting point. while impure substances melt over  
a range of temp.

### Crystallization

\* Conditions for <sup>selecting</sup> picking a solvent;

- It must be soluble in hot <sup>Phase</sup> not cold Phase.

\* Which of these is ~~sol~~ suitable;

| <del>Benzene</del> | Hot Phase | Cold Phase. |
|--------------------|-----------|-------------|
| Benzene            | ✓         | x           |
| Ethanol            | ✓         | ✓           |
| Toluene            | x         | x           |
| Fluorine           | ✓         | ✓           |

Benzene is the best solvent.

BELOVED



## Below

\* Common solvents used in recrystallization is acetone, water or animal charcoal

\* We use activated charcoal to remove <sup>coloured</sup> impurities.

### Filtration

- used to separate solids from liquids using filter paper
- Vacuum filtration used for cold solution and <sup>for</sup> rapid separation of crystals

~~Rapid filtration~~ Gravity filtration is used for hot solution & to prevent premature crystallization.

### Sublimation

- A substance from solid directly into gas. the reverse is deposition, i.e from gas directly to solid
- We use it to purify volatile solids.
- There's decomposition.

### Drying Agents and Desiccator

- Desiccator is used to provide moisture free environment

Examples of drying agent — Silica gel, <sup>Anhydrous</sup> Calcium chloride, concentrated  $H_2SO_4$ , Anhydrous Copper sulphate  $CuSO_4$ , <sup>Anhydrous</sup>  $MgSO_4$ ,  $Na_2SO_4$ , Sodium sulphate, ~~anhydrous~~  $CaCl_2$ , Quick lime  $CaO$ , removes  $CO_2$



BELOWED !

- Desiccator is used for drying

### \* Chromatography

- Thin Layer Chromatography (TLC)

we use it to check for purity.

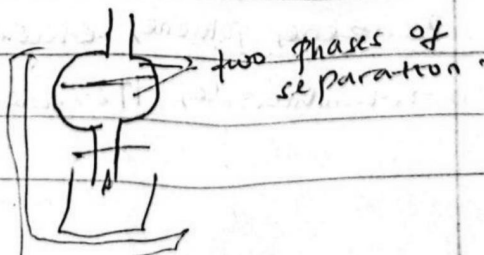
- It has a mobile phase & stationary phase.
- Different polarity between mobile & stationary phase.
- Stationary phase — silica gel, alumina,
- Retention factor  $R_f = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}}$

### Paper Chromatography

→ The one with low affinity with the silica gel will come out faster. The principle of TLC is partition between the stationary phase and mobile phase.

\* We use separating funnel to separate immiscible liquids in extraction.

\* The solvent used must be immiscible with water. E.g. Chloroform, Benzene.



BELOVEDSolubility test.

- "Like dissolves like"
- "Polar solutes dissolve in polar solvents"
- "non-polar <sup>solute</sup> ~~solvent~~ dissolves in non-polar solvents."
- Hydrocarbons are insoluble in water
- <sup>some</sup> Amines are soluble in water due to
- " hydrogen bonding.
- Solubility decreases in increase in chain length.
- Solubility test helps in functional group identification.

The ~~same~~ in solubility like dissolves like.

- List examples of drying agent
  - Which of the following will dissolve in water?
- Ⓐ Toluene Ⓑ Benzene Ⓒ Hexane Ⓓ none of the above

\* NaCl dissolves in water.

- Large Alkanes,

\* What types of substance are more soluble in water?

- Ⓐ Aromatic H<sub>2</sub>C Ⓑ non-polar Ⓒ

\* Insoluble organic compounds in water may dissolve in Benzene, Toluene, Ether, Hexane, Carbon tetrachloride, Hexane.

\* Hydrolysis, Hydration, dehydration!

\* Which substance is ~~now~~ least soluble in water?

Ⓐ propanol Ⓑ ethanol Ⓒ Hexanol Ⓓ Heptanol

\* Solubility test distinguishes between polar and non-polar

Hydrolysis — breaking a bond using water

Hydration — Addition of water

• What is the accurate method of determining boiling point?

Ⓐ capillary method Ⓑ distillation method

• What is the atmospheric pressure at ~~a~~ STP.

760 mmHg

## Paper 10 of 10 — BAMBOY 100L 2nd semCHE 104 (BAMBOY)separation in organicchemistry- SNOW DON-Copy.pdf

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### SEPARATION IN ORGANIC CHEMISTRY

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#### ABSTRACT

Separation methods used in organic chemistry are briefly reviewed, emphasis being placed on aspects that are frequently ignored, and special attention being paid to the analytical applications of separation. The relevance of physical, inorganic and analytical chemistry for organic chemistry is stressed.

The term 'separation' can be used in analytical chemistry in both a literal and figurative sense (an idea that is due to Professor Čuťa<sup>1</sup>). In figurative separation, use is made of some difference in properties of the individual components of a mixture in order to determine one of them without physical separation from the others. In this sense, a selective titration or a spectrophotometric measurement may be a 'separation' in that no other species is measured together with the one of interest. The object of this paper is to present a survey of both literal and figurative methods of separation in organic chemistry, particular attention being paid to certain features which are often forgotten in application of the methods.

#### PHYSICAL SEPARATION

Physical separation may be used for isolation, purification, identification and determination of organic compounds. It is, of course, generally a two-phase process, and can be classified into solid-liquid, liquid-liquid, gas-liquid and gas-solid systems. An exception is electrophoresis, which takes place in a single phase. Applications of physical separation may cut across these categories—for example, phase titrations may result in separation of a solid or a liquid phase from the initial liquid phase. Similarly, some of the factors associated with physical separation have implications for figurative separation.

##### Solid-liquid systems

A major application of these is in precipitation methods, which can be taken to include crystallization techniques, and they will mainly be used for isolation and purification. It is instructive to consider what degree of purity is likely to be required. If we take an organic compound containing 60 per cent carbon, then for an analysis (assumed to be made without other errors) to fall within the accepted limits of  $\pm 0.3$  per cent carbon the compound must contain not more than 0.5 per cent of a non-carbonaceous compound, i.e. it must be at least 99.5 per cent pure. If it is contaminated with carbonaceous material, however, a lower level of purity may occur without detection by analysis. Thus such a

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compound recrystallized from glacial acetic acid (40 per cent carbon) could contain 1.5 per cent of the solvent and still show a 'correct' result; for benzene (92 per cent carbon) as solvent, one per cent of it could escape detection. This is an oversimplification, because the addition of experimental error to the bias introduced by the impurity would be as likely as not to make the result fall outside the limits of error, but there is still a 50 per cent chance that the impurity would escape detection. Only if the amount of impurity were doubled would it be fairly certain to be found. It follows that for application of classical microanalysis, a purity of 99.5 per cent or better is required. We shall return to this point later. A further complication is the possibility of segregation of impurity so that the sample taken for analysis may contain more or less than the average amount. The smaller the sample, the less the chance of choosing a more heavily contaminated portion, but the more disastrous the consequences if such a portion is chosen.

There is a considerable body of information and instruction available on crystallization of organic compounds<sup>2</sup>, with emphasis on the art of inducing crystallization to occur when it does not want to. There is sometimes a feeling that the art is dying and science left out, and that some organic chemists, especially research students, do not think hard enough about what they are doing. For example, although the nature of the precipitation process and of contamination have been extensively studied in inorganic and physical chemistry, the results are not always applied as they might be in organic chemistry. It is ironic to reflect that the technique of precipitation from homogeneous solution was first used in organic chemistry, for example in precipitation from aqueous alcohol by addition of water to the pure alcohol solution until a precipitate appears (i.e. a phase titration, see below) followed by warming to redissolve the precipitate and then cooling until crystals appear, yet an organic chemist has been heard to observe that a compound *must* be pure because it was obtained by precipitation (irrespective of the conditions used). Very often, the same solvent system is used for repeated recrystallization in an attempt at purification, even though the solvent itself may be strongly adsorbed (especially glacial acetic acid) or the adsorption isotherm for an impurity favours extensive co-precipitation. The reason usually given is that the compound is too soluble in other solvents; for analytical purposes, however, it is purity that counts, not the yield, and recovery of a small fraction of pure material could be achieved from almost any solvent, and the remainder could be recovered by evaporation of solvent. Glacial acetic acid is notoriously hard to remove if it has been used as solvent, but recrystallization from another solvent will often reduce the contamination to an acceptable level. Mixed solvent systems for precipitation from homogeneous medium might also be explored rather more extensively. Another possibility for improving the purity of an analytical sample is due to the late Professor Gordon<sup>3</sup>, and consists of careful control of the crystallization process and separation of that fraction which will be least contaminated. For example, co-precipitation may occur extensively on the first portion of precipitate formed, or may be delayed until precipitation is almost complete (*Figure 1*). True, this is difficult to realize in practice, but it might be worth exploration.

The second major application is in chromatography, whether by adsorption on columns or thin layers, by ion-exchange, or by gel-filtration. Here it may

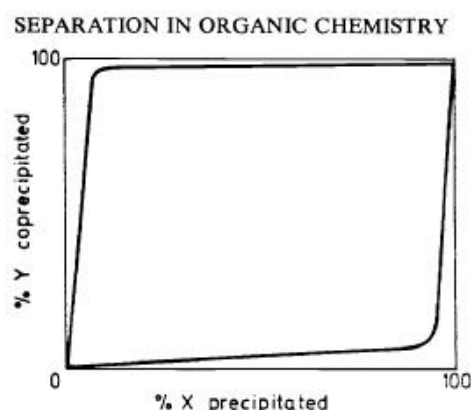


Figure 1. Diagrammatic representation of two extreme types of co-precipitation of Y with X. Collection of only a fraction of the precipitate will give a product that is less contaminated.

happen that a large volume of solvent is necessary for elution of a small amount of compound, followed by evaporation of the solvent. The solvent may contain sufficient impurity itself, or extract enough material from a column of support, to leave a significant amount of contamination behind when it is removed, and it is certainly desirable to use a further purification process before use of the material for analysis. The presence of 1 p.p.m. of impurity in 500 ml of solvent yields  $\sim 500 \mu\text{g}$ —ten per cent of a 5 mg recovery! The possibility of specific interaction with the solid phase must not be forgotten—*isomerization, cyclization, oxidation, condensation etc.*, and even total destruction, may occur. All too often a single and unsuitable solvent is applied to develop a chromatogram and the resultant single spot of unresolved mixture is regarded as evidence that a single substance is present. At least two solvents of differing polarity should be tried, and the starting materials and likely byproducts tested individually under the same conditions.

Zone refining<sup>4</sup> is a technique for purification that is perhaps not used as much as it might be, probably because the usual scale of working in organic chemistry is too small. Some interesting possibilities have been suggested by recent work on organometallic complexes, notably metal  $\beta$ -diketonates<sup>5</sup>, and it is interesting to speculate on the possibilities of deliberate addition of ions to remove impurities that can form metal complexes.

#### Liquid-liquid systems

Solvent extraction is the obvious example of these systems, with its variants such as countercurrent extraction and partition chromatography. Although adjustment of acidity has long been exploited for separation of compounds containing suitable functional groups, and extraction of organometallic compounds is commonplace in analysis, little seems to have been done to utilize metal complexes for isolation and purification of organic compounds containing suitable ligand groups and atoms. A similar approach that *has* been used is the formation of extractable ion-association compounds between large cations such as tris (1,10-phenanthroline) iron(II) or tetraphenylarsonium and organic acid anions<sup>6</sup>. Extraction of small amounts of impurity is fairly easily dealt with by washing with the initial solvent. If A is 99 per cent extracted into C from D,



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but B is only one per cent extracted, then washing the extract with pure D will lower the concentration of B in C to 0.01 per cent of its original concentration in C.

Possibilities for exploration include the use of low-melting solids for extraction at temperatures above their melting point, followed by cooling and separation of the solid extract<sup>7</sup>, and the exploitation of polarity by use of mixed solvents (cf. the mixotropic solvent series in partition chromatography).

Phase titrations are an example of two-phase systems used in analysis, and can be either liquid-liquid or liquid-solid systems. Such methods are laborious because they require a considerable amount of calibration, and are restricted in scope to simple mixtures of known qualitative composition, but can be very useful for routine analysis of suitable systems<sup>8,9</sup>.

**Gas-liquid systems**

Partition gas chromatography has become of much greater importance than distillation, and when coupled with other techniques such as mass spectrometry, infra-red spectroscopy etc, offers a very powerful tool indeed. For some purposes, however, distillation may still be the better method, especially if relatively involatile compounds are to be separated, when distillation at reduced pressure may solve problems of thermal instability.

As remarked above, chromatography of any kind, as a test of purity, should be used with caution, and the behaviour of likely impurities and byproducts and of the starting materials should be tested under the same conditions as the test compound; alternatively, but perhaps less satisfactorily, the test sample alone should be chromatographed on more than one stationary phase. In preparative gas chromatography, the possibility of bleed of stationary phase from the column must be considered, and the fractions collected may need further purification.

**Gas-solid systems**

Adsorption chromatography and sublimation come under this heading. Interaction with the absorbent must be watched for in the former, and in the latter attention must be paid to the possibility of thermal decomposition giving volatile products that may be condensed along with the sublimate. It should also be remembered that hygroscopic substances can still sorb water from the atmosphere after sublimation, unless the operation is done in a dry atmosphere.

**Mass spectrometry**

Although mainly used as a tool for analysis and for structural investigation, the mass spectrometer could in principle be used for isolation of certain species, though the process would be rather expensive, limited in scope and wasteful because of fragmentation of the parent compound. A major analytical application is the detection of impurities; if the peaks corresponding to these are well separated from the main spectrum, detection limits of the order of parts per million may be attained. It is easier to detect impurities if they have molecular

## SEPARATION IN ORGANIC CHEMISTRY

weights higher than that of the major component; lower homologues may not always be detected as impurities unless present in relatively large amount. The substance to be determined must have a certain degree of volatility, of course, but even involatile compounds can be examined if flash pyrolysis or a suitable chemical reaction can provide volatile products, and the presence of volatile impurities is easily detected. A more recent use of mass spectrometry, and one that has gained acceptance by the journals, is for determination of elemental composition of compounds.

## FIGURATIVE SEPARATION

Figurative separation involves a means of measurement which will also provide differentiation between components of a mixture, and this usually means use of spectroscopy or electrochemical methods.

**Spectroscopic methods**

Ultra-violet, visible region and infra-red absorption spectroscopy are well tried techniques in organic chemistry, and problems really arise only when mixtures with overlapping spectra are to be analysed, or if impurities contribute a background spectrum. Traditionally such problems were solved by use of simultaneous equations applied to a series of measurements at suitable wavelengths, but naturally errors were propagated rather rapidly as the number of measured components increased. The advent of the computer and the cathode-ray oscillograph has led to the development of spectrum simulators in which a series of spectra can be varied at will and combined until they match the observed spectrum, but it is still necessary to have a good idea of how many contributory spectra are important ones, as otherwise a family of solutions might be possible. One approach to the problem is to treat the individual absorption peaks as Gaussian in shape and to analyse the peaks one by one across the spectrum. Again a certain amount of caution is necessary in interpretation, lest an impurity should have a single peak that coincides with a peak from the major component, and so not only escape detection itself but cause an error in the determination of the other component. It is frequently assumed that if a background spectrum appears superimposed on that of the substance examined, all that is needed is to compare the spectrum with that of pure substance and assume that the contribution of the impurity to the peak of interest is a linear function of the wavelength. This assumption may not always be justified, however; and various mathematical methods have been proposed for resolving the problem, usually by a trial and error fitting of other simple mathematical functions of absorbance with respect to wavelength<sup>10</sup>.

A further problem arises in connection with the sensitivity of the methods as a means of detecting impurities. The relationship between absorbance and concentration is a logarithmic one, and the reading errors are minimal over a relatively short portion of the absorbance scale. Even when differential methods are used, the experimental error in determining the concentration is likely to be a few parts per thousand, and the method will not detect amounts of impurity much below say 0.5–1 per cent.



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Fluorescence spectroscopy is a much more potent tool for detection of impurities, since the intensity of the light emitted is a function of the intensity of the exciting radiation, and very small amounts of fluorescing impurity (in the ng region) can be detected in a non-fluorescent matrix. A somewhat similar potential lies in laser-excited emission of infra-red radiation, since a very intense radiation can be realized and a long excitation pathlength used<sup>11</sup>. There are also several possibilities for improving the selectivity of fluorescence methods. Enhancement or quenching of fluorescence can be achieved by change of solvent medium, change of pH, coupling and redox reactions etc, and both the excitation and emission spectra can be used for identification purposes<sup>12</sup>.

Nuclear magnetic resonance spectroscopy is now a favourite tool for elucidation of structure, especially the identification of functional groups and determination of the number of them present. The snag is that while the method is invaluable for obtaining information about the major component(s) of a mixture, it is not sufficiently sensitive to yield information about minor or trace components (i.e. present in amounts below about two per cent). Consequently it may happen that a research worker prepares a compound of sufficient purity to yield the n.m.r. spectrum of the major component only, decides that the compound is the one intended, and submits the unpurified product for elemental analysis only to find that the results do not agree with the expected ones within experimental error. The reason is obvious. As pointed out above, the minimum purity required for a sample suitable for analysis is at least 99.5 per cent, and although the research worker knows that he has made a particular compound, the analyst must perforce analyse the material supplied as a sample, and if this happens to be a mixture the results will be number averages derived from the relative amounts and compositions of the components of that mixture. In such a case both the analyst and the research worker are 'right'—the one in providing analytical results that are not those expected, and the other in having correctly identified the main component of the sample.

**Electrochemical methods**

Polarography and its ancillary techniques such as oscillographic polarography, voltammetry etc, are useful in many applications in organic chemistry and analysis. The anodic and cathodic reactions possible with many functional groups provide considerable scope for differentiation of components of mixtures, and the sensitivity of the methods is very useful in trace analysis.

**Differential titration**

By suitable methods, such as spectroscopic or conductometric determination of the end-points, it is possible to titrate a strong and a weak acid in succession, and mixtures of such organic acids can be analysed. The strength of acids which are too weak for normal titration can be increased by various means, such as change of solvent, or use of a metal ion to form a complex with accompanying release of protons. In some circumstances differential redox reactions may be possible.

## SEPARATION IN ORGANIC CHEMISTRY

## COMBINATION OF METHODS

When one or other of the two types of separation cannot alone resolve the composition of a mixture, a judicious combination of the two may provide the answer to the problem. For example, solvent extraction of a metal complex of an organic compound has been followed by atomic-absorption determination of the metal in order to determine the organic moiety of the complex<sup>13</sup>. Again, the products of electrolysis at the dropping mercury cathode have been identified by submicroanalysis and gas-liquid chromatography<sup>14</sup>. Selective extraction or chromatographic separation is often used to simplify complex mixtures before spectroscopic or other determinative methods are applied. Precipitation is often used to remove unwanted material before working up the main product, or using an analytical procedure with which the foreign material would interfere. Many reactions in which organic reagents are used for inorganic analysis can be turned round and used the other way, as Feigl has repeatedly shown.

To sum up, while organic chemistry has often provided fruitful ideas for inorganic and analytical chemistry, there are many respects in which it may learn from these branches of chemistry, and two-way traffic of this kind can only be of benefit to chemistry as a whole.

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